Ultrasonics 159 (2026) 107852

Contents lists available at ScienceDirect

Ultrasonics

journal homepage: www.elsevier.com/locate/ultras

S0 Lamb wave scattering in plate structures using physics-enhanced TransUNet

Linfeng Wang [a], Hongyan Zhang [a], Zhen Zhang [a], He Sun [a], Xiao Ying [a], Jing Hou [b], Jian Li [a], Zhoumo Zeng [a], Yang Liu [a,c,*]

[a] State Key Laboratory of Precision Measurement Technology and Instruments, Tianjin University, Tianjin 300072, China
[b] CNOOC Research Institute Co., Ltd., Beijing 100028, China
[c] International Institute for Innovative Design and Intelligent Manufacturing of Tianjin University in Zhejiang, Shaoxing 330100, China

ARTICLE INFO

Keywords:
Lamb wave scattering
Physical constrains
Irregular defects
Wave propagation

ABSTRACT

Lamb wave scattering offers valuable insights into material properties, defect characteristics, and wave propagation behaviors, making it a prominent research topic in the fields of non-destructive testing and structural health monitoring. Analytical methods, while offering theoretical analysis, are often limited to simple defect geometries and homogeneous media. Numerical methods such as the boundary element method and hybrid techniques can handle complex structures but suffer from computational costs, particularly at high frequencies due to the need for fine discretization. To overcome these challenges, this study proposes a physics-enhanced TransUNet (PTUNet) for solving the scattered wavefield of S0 Lamb waves in plate structures with irregular defects. PTUNet combines the local feature extraction capability with the global modeling capacity while employing finite difference approach to incorporate the Kirchhoff-Love plate theory as a physical constraint. The effectiveness of the proposed method is validated through numerical simulations on random defects and experimental measurements using a scanning laser Doppler vibrometer on a 1.5 mm-thick aluminum plate. The results demonstrate that PTUNet accurately predicts wave propagation, mode conversion, and intricate scattering characteristics, achieving reasonable agreement with both finite element simulations and experimental observations. PTUNet presents a promising approach for wavefield modeling and defect characterization, with potential applications in acoustic scattering problems.

1. Introduction

In guided wave propagation, scattering refers to the alteration of wave paths and characteristics when encountering structural discontinuities, defects, or geometric variations. Scattering is both a fundamental wave phenomenon in physics and a critical area of research in nondestructive testing (NDT). The analysis of scattering waves provides valuable insights into structural defects, material properties, and stress distribution, facilitating effective structural health monitoring and evaluation. In comparison to traditional inspection techniques, guided wave scattering offers significant advantages in NDT, especially for large or complex structures. First, guided wave scattering provides global information with strong penetration capability, allowing for the effective detection of defects in large-area or thick-walled structures—something that conventional localized inspection techniques often struggle to achieve [1]. Second, scattering exhibits high sensitivity to small defects, enabling the detection of hidden flaws that may not be identifiable with traditional methods [2]. Furthermore, analyzing variations in the scattering waveforms provides multidimensional insights into the structural state, enhancing defect localization accuracy and detection efficiency [3]. Finally, guided wave scattering demonstrates remarkable adaptability, making it applicable across a wide range of materials and structures [4]. This is particularly beneficial for inspecting complex geometries or areas that are difficult to access, where traditional methods may have limitations. Through a systematic study of guided wave scattering mechanisms, researchers can develop a more accurate theoretical foundation and practical guidelines for assessing the reliability of materials and structures, thus advancing the field of NDT.

Over the past few decades, extensive research has been conducted on the scattering of guided waves interacting with defects in plate structures. Fundamental investigations into guided waves in elastic solids can

* Corresponding author at: State Key Laboratory of Precision Measuring Technology and Instruments, Tianjin University, Tianjin 300072, China.
E-mail address: ultrasonicslab@tju.edu.cn (Y. Liu).

https://doi.org/10.1016/j.ultras.2025.107852
Received 7 May 2025; Received in revised form 25 August 2025; Accepted 6 October 2025
Available online 8 October 2025
0041-624X/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

be found in the works of Achenbach [5] and Auld [6]. A notable study by Diligent et al. [7] focused on the scattering of incident S0 Lamb waves from circular through-holes in plates. By employing 3D theory and expanding the Lamb and SH wave fields using Bessel functions, the authors applied a projection method incorporating stress and displacement boundary conditions to determine the expansion coefficients. Simulations and experimental validation revealed that scattering coefficients increase with hole diameter and decrease with distance from the hole. Building on this work, Grahn [8] and Moreau et al. [9] extended the study of 3D scattering of S0 Lamb waves to partially through-thickness circular cavities and irregularly shaped blind holes. It is noteworthy that [7] and [8] selected sine functions as projection functions, which were found to be unstable with slow convergence rates. In contrast, [9] adopted displacement components of Lamb waves as projection functions, significantly improving convergence speed. While precise 3D theory remains effective across both low and high frequencies, its computational complexity is a major limitation. To address this challenge, researchers have explored plate theories as an alternative for modeling low-frequency scattering. Specifically, Poisson theory and Kirchhoff theory have been employed to characterize the extensional and flexural motions of guided waves in plates. Grahn [8] formulated the scattering problem using Poisson-Kirchhoff theory, converting displacement, stress, and moment boundary conditions into matrix form, and solving for the unknown expansion coefficients. This approach allowed for the calculation of far-field amplitudes of scattered S0, SH0, and A0 waves resulting from S0 Lamb wave incidence on a circular blind hole. A comparison with 3D theory demonstrated that Poisson-Kirchhoff theory provides high accuracy at low-frequencies. However, as the incident frequency increases, its effectiveness diminishes. To overcome this limitation, Kirchhoff theory was replaced by Mindlin theory, which maintains similar computational efficiency while offering a more accurate representation of flexural wave propagation in plates. Cegla et al. [10] employed Poisson-Mindlin theory to model guided wave scattering at non-symmetric blind holes in isotropic plates. Comparisons with 3D analytical solutions and acoustic field scanning experiments confirmed that Mindlin theory outperforms Kirchhoff theory in high-frequency scenarios. Further extending this approach, Sedaghati et al. [11] developed an S0 Lamb wave scattering model for circular cavities in isotropic plates, capable of handling both symmetric and asymmetric single- and double-sided blind holes. Similar studies can be found in [12,13]. In recent years, Poddar and Giurgiutiu [14] introduced a novel analytical method, Complex Modes Expansion with Vector Projection (CMEP), for solving Lamb wave scattering problems in plates. Unlike previous projection methods, CMEP enhances accuracy by projecting stress boundary conditions onto the conjugate of displacement modes and vice versa. This formulation leverages power flow associated with Lamb wave modes, transforming stress and displacement equations into power equations, leading to faster convergence. This method was applied to analyze the scattering behavior of horizontal cracks, disbands [14], steps [15,16], and horizontal cracks in a stiffened plate [17,18]. Moreover, Shen and Giurgiutiu [19] developed WFR-2D, an analytical prediction tool based on precise 2D Lamb wave modeling, which facilitates ultrasonic guided wave simulation in damaged structures. Research has also extended beyond individual defects to explore guided wave scattering in plates with multiple defects, as demonstrated in [20,21]. Extensive analytical research has been conducted on scattering problems in plates. However, the mathematical complexity of analytical methods has restricted most studies to simple defect geometries, primarily focusing on the angular-dependent energy distribution characteristics of the scattered wavefield. As a result, current analytical approaches have yet to fully capture time-domain scattered signals or reconstruct complete scattered wavefields.

In addition to analytical solutions, numerical methods such as the finite element method (FEM), boundary element method (BEM), and spectral element method (SEM) have been widely used to study wave scattering [22–24]. These methods often offer advantages when dealing with complex scatterers, where analytical approaches may become intractable. To improve computational efficiency, especially in high-frequency scattering problems with complex defect geometries, researchers have developed hybrid methods that combine FEM, BEM, and mode expansion. For instance, Cho et al. [25] employed a hybrid BEM approach that integrates boundary integral equations with mode expansion techniques to numerically determine reflection and transmission coefficients for different defect conditions. However, numerical methods inevitably involve a trade-off between solution accuracy and computational cost, particularly regarding mesh resolution. This challenge intensifies in high-frequency problems, which demand finer meshes and significantly increase computational complexity. A practical alternative is the semi-analytical finite element method (SAFE), which allows for 3D scattering analysis using a 2D guided wave framework. For example, Maruyama et al. [26] employed a hybrid SAFE-BEM method to investigate wave scattering in isotropic, linear elastic plates with finite width. Despite their benefits, these methods still require complex analytical formulations within the FE framework, which poses challenges in terms of user-friendliness and practical applicability. Further improvements are necessary to improve accessibility and computational efficiency for broader applications in guided wave scattering analysis.

With advancements in deep learning, neural networks have become a powerful and versatile tool for wave scattering analysis. In recent years, researchers have increasingly applied deep learning to wave scattering, introducing innovative methodologies and improving computational efficiency. For example, Shao et al. [27] proposed a multi-task convolutional neural network to investigate Lamb wave scattering in composite plates. By incorporating the Born approximation principle to generate a multi-damage dataset, this approach enhanced the accuracy of damage quantification. Ji et al. [28] introduced an asymmetric autoencoder architecture for the full wavefield reconstruction of Lamb waves in carbon fiber-reinforced plastic (CFRP) structures, achieving high reconstruction accuracy even under high sparsity conditions and exhibiting strong generalization across different damage scenarios. Beyond these studies, deep learning has been effectively applied to wave propagation and damage detection, showcasing its broad potential in this field [29–31]. Despite their success, data-driven deep learning models rely heavily on large, high-quality training datasets. The scarcity or imbalance of labeled data across different damage types can severely affect the generalization and predictive accuracy of these models. Theoretically, the required training data grows exponentially with the input dimensionality, yet in real-world engineering applications, acquiring such extensive datasets is often impractical. This data limitation has become a major challenge, restricting the broader adoption of deep learning in practical wave scattering problems.

To overcome these limitations, physics-informed neural networks (PINNs) and their variants [32–35] have emerged as a promising alternative, attracting significant attention across various research domains. Unlike conventional data-driven models, PINNs incorporate fundamental physical laws such as governing equations and boundary conditions directly into the loss function using automatic differentiation. This approach ensures that the model learns from data while strictly adhering to physical constraints, improving both accuracy and generalization. Compared to numerical methods, PINNs eliminate the need for meshing, mitigating the trade-off between computational cost and solution accuracy. This advantage makes them particularly effective for complex geometries and high-dimensional problems. Furthermore, PINNs require significantly fewer labeled datasets, as they can be trained solely based on governing equations, ensuring physically consistent predictions even in out-of-distribution scenarios. PINNs have already shown great promise in ultrasonic wave scattering applications. In previous work, we have successfully employed PINN frameworks to model ultrasonic wave propagation in plates containing single and multiple defects, validating their effectiveness in solving complex scattering problems [36]. Sun et al. [37] utilized a high-frequency shear horizontal (SH) guided wave-based PINN for microcrack quantification,

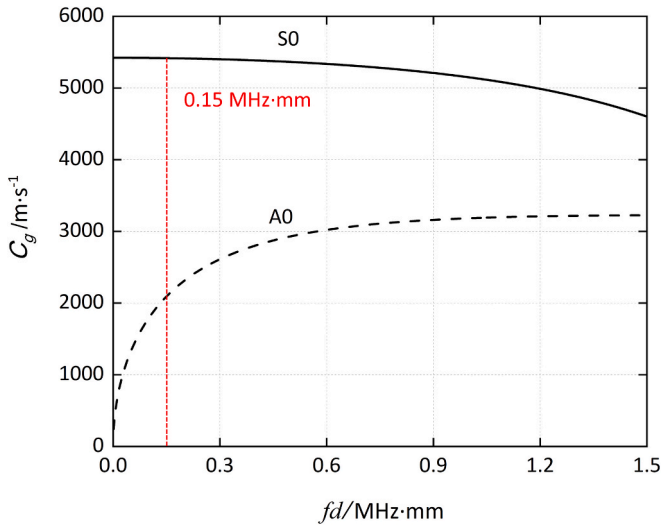


Fig. 1. Group velocity dispersion curve of Lamb waves.

reducing quantification error by over 80 % compared to conventional methods. Zhou et al. [38] applied PINNs to extract undamaged wavefields from damage-affected wavefields, enabling precise defect identification using signal processing techniques. Go et al. [39] employed PINNs to analyze elastoacoustic interactions in perforated and cracked plates, achieving millimeter-scale accuracy in stress and displacement estimations. These studies collectively highlight the effectiveness of PINNs in wavefield reconstruction, defect quantification, and damage localization. However, research leveraging this approach for Lamb wave scattered wavefield reconstruction remains notably scarce.

Unlike conventional neural networks which are typically designed to approximate finite-dimensional input–output mappings (i.e., point-to-point or vector-to-vector relations), neural operator methods aim to directly learn mappings between functions. Representative approaches include DeepONet and Fourier Neural Operator (FNO). DeepONet employs a branch–trunk architecture, where the branch encodes the sampled values of the input function and the trunk encodes the output coordinates, with their inner product approximating the operator [40,41]. However, for large-scale spatiotemporal wavefields, it requires explicit (x, y, t) grids and dense sampling, leading to high dimensionality and training cost. FNO, on the other hand, leverages Fourier transforms to perform global convolutions in the spectral domain, making it particularly powerful for capturing large-scale propagation [42,43]. Yet, being data-driven, FNO does not strictly enforce governing PDEs, which can reduce physical consistency in local scattering. Prior studies have also suggested that adding PDE constraints can improve generalization in more challenging settings [44].

This paper proposes a physics-enhanced TransUNet (PTUNet) framework for solving the scattered wavefields of S0 Lamb wave in plate structures by integrating data-driven learning with physical constraints. Previous research has demonstrated the effectiveness of incorporating physical priors into data-driven models to improve prediction accuracy and generalization ability [45]. We embed Kirchhoff-Love plate theory into the TransUNet framework using finite difference (FD) technique as a physical constraint within the loss function. This integration enables PTUNet to jointly leverage data-driven learning and fundamental physical laws. Compared to purely data-driven approaches, the incorporation of physical constraints significantly reduces the reliance on large-scale labeled datasets, prevents unphysical solutions, and enhances model generalization. It should be emphasized that the "physics-enhanced" paradigm is not intended to replace other methods, but rather as an integration of physical priors into deep models. In this study, we conduct FE simulations to model the propagation of S0 Lamb waves incident on irregular defects in 3D plate structures, constructing a dataset containing out-of-plane displacement information. Numerical simulations validate the effectiveness of the proposed method in improving prediction accuracy and enhancing adaptability for varying defect geometries. Furthermore, scanning laser Doppler vibrometer (SLDV) experiments are conducted to obtain high-resolution wavefield measurements of a 1.5 mm-thick aluminium plate containing an irregular defect. Experimental results demonstrate that the predictions of PTUNet achieve reasonable consistency with FE simulations and experimental measurements. These findings highlight PTUNet as a robust and efficient framework for guided wave scattering analysis, offering a promising solution for NDT and structural health monitoring applications.

2. Methodology

2.1. Propagation of Lamb waves in isotropic and homogeneous plates

2.1.1. 3D theory

The propagation theory of Lamb modes and shear horizontally polarized (SH) modes in isotropic, homogeneous, linearly elastic, and infinitely extended plates has been discussed in detail in [8,46]. Fig. 1 depicts the dispersion curve of Lamb waves in a 1.5 mm aluminium plate. The low-frequency S0 Lamb wave, with high group velocity and low attenuation, effectively detects deep defects and experiences minimal attenuation in fluid media, making it ideal for long-range NDT of plate structures. Although the lowest frequency is preferred to avoid dispersion, a balance between resolution and frequency is often necessary in practice. The frequency-thickness product is typically limited to 1.6 MHz·mm to avoid the A1 mode cut-off frequency [7]. This study focuses on frequencies of 150 kHz·mm, where the S0 mode exhibits negligible dispersion.

The material properties of the plate are characterized by Young's modulus E and Poisson's ratio ν, or alternatively by the Lamé constants λ and μ, with the density denoted as ρ. The thickness of the plate is $2h$. The following derivation is formulated under the standard assumptions of time-harmonic fields, linear elasticity, and plate isotropy. Assuming the incident wave is a plane far-field S0 wave and the time factor $\exp(-iwt)$ is omitted under the time-harmonic condition. In a cylindrical coordinate system, the wave motion of Lamb modes can be expressed as:

$$u_r^n = \frac{1}{k_n} V_n(z) \frac{\partial \phi}{\partial r}(r,\theta),\ u_\theta^n = \frac{1}{k_n} V_n(z) \frac{\partial \phi}{\partial \theta}(r,\theta),\ u_z^n = W_n(z)\phi(r,\theta). \tag{1}$$

The expressions of $V_n(z)$ and $W_n(z)$ for symmetric and antisymmetric modes in Lamb waves are provided in [46]. The scalar potential function $\phi(r,\theta)$ satisfies the Helmholtz equation:

$$\frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r}\frac{\partial \phi}{\partial r} + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2} + k_n^2 \phi = 0. \tag{2}$$

The wavenumbers k_n for symmetric and antisymmetric modes are given from the Lamb-Rayleigh equation:

$$\begin{aligned} &\frac{\tan(qh)}{\tan(ph)} = -\frac{4k^2 pq}{(q^2-k^2)^2}, \text{ for symmetric modes}, \\ &\frac{\tan(qh)}{\tan(ph)} = -\frac{(q^2-k^2)^2}{4k^2 pq}, \text{ for antisymmetric modes}, \end{aligned} \tag{3}$$

where

$$q^2 = \frac{\omega^2}{c_T^2} - k^2,\ p^2 = \frac{\omega^2}{c_L^2} - k^2. \tag{4}$$

The quantities c_T and c_L represent the velocities of shear waves and longitudinal bulk waves, respectively, while ω denotes the angular frequency. Now, we assume that a solution for the potential function $\phi(r,\theta)$, satisfying Eq. (2), can be expressed as:

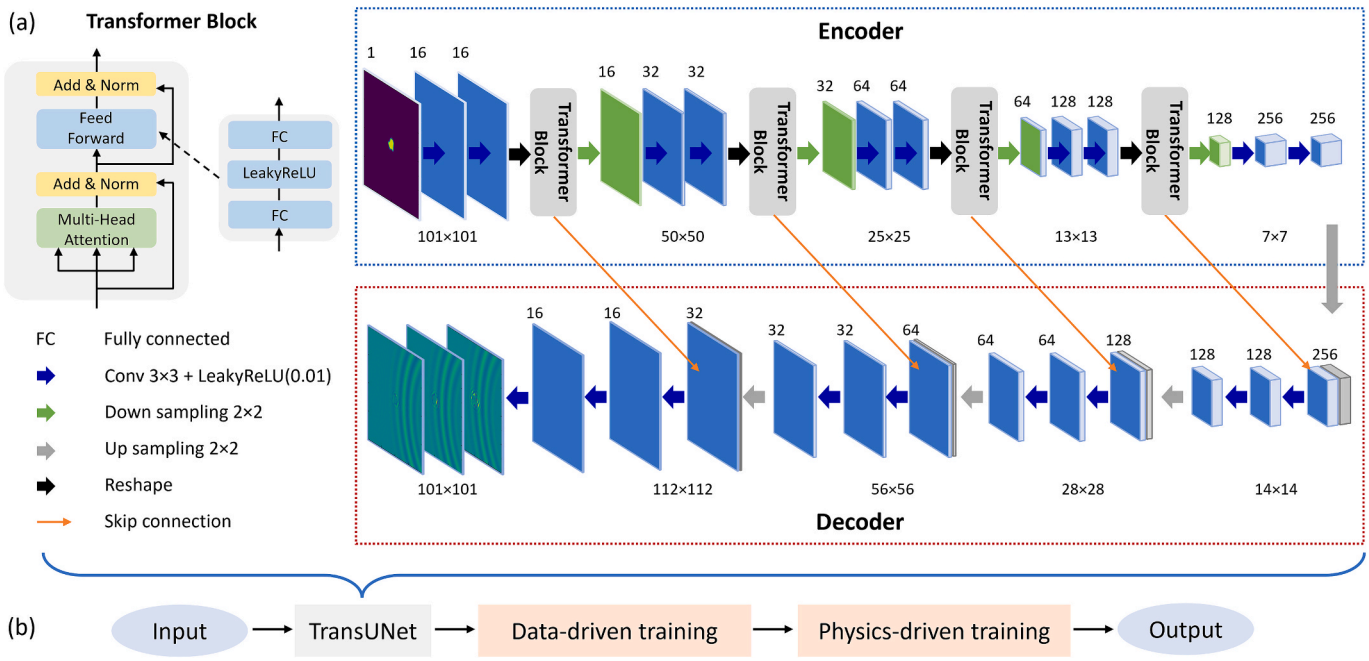


Fig. 2. The architecture of PTUNet.

$$\phi(r,\theta) = \Phi(r)e^{im\theta}, \tag{5}$$

where $\Phi(r)$ is a solution of Bessel's differential equation and satisfies

$$\frac{d^2\Phi}{dr^2} + \frac{1}{r}\frac{d\Phi}{dr} + \left(k_n^2 - \frac{m^2}{r^2}\right)\Phi = 0. \tag{6}$$

The wavefield is divided by the boundary of the defect into two regions: the outer wavefield (denoted by "$>$") and the inner wavefield (denoted by "$<$"). For outgoing waves of the Lamb modes, the solutions of Eq. (2) are:

$$\begin{aligned} \phi_{mn}^{>}(r,\theta) &= H_m\left(k_n^{Sh}r\right)e^{im\theta}, \textit{ for symmetric modes},\\ \phi_{mn}^{<}(r,\theta) &= H_m\left(k_n^{Ah}r\right)e^{im\theta}, \textit{ for antisymmetric modes}, \end{aligned} \tag{7}$$

where H_m is the first type of Hankel functions. The wavenumbers k_n^{Sh} and k_n^{Ah} are obtained from Eqs. (3) and (4), representing symmetric and antisymmetric modes, respectively. The half plate thickness is h. Assume that the remaining thickness of the plate inside the defect is $2b$. Then in the inner region, the solutions of Eq. (2) are:

$$\begin{aligned} \phi_{mn}^{<}(r,\theta) &= J_m\left(k_n^{Sb}r\right)e^{im\theta}, \textit{ for symmetric modes},\\ \phi_{mn}^{<}(r,\theta) &= J_m\left(k_n^{Ab}r\right)e^{im\theta}, \textit{ for antisymmetric modes}, \end{aligned} \tag{8}$$

where J_m is the first type of Bessel functions.

The scattering process also gives rise to SH modes, which can likewise be expressed using a scalar potential function $\psi(r,\theta)$ and a thickness-dependent function $U_n(z)$:

$$u_r^n = \frac{1}{l_n}U_n(z)\frac{1}{r}\frac{\partial\psi}{\partial\theta}(r,\theta),\ u_\theta^n = -\frac{1}{l_n}U_n(z)\frac{\partial\psi}{\partial r}(r,\theta),\ u_z^n = 0, \tag{9}$$

where $\psi(r,\theta)$ also satisfies the Helmholtz equation, and l_n corresponds to the wavenumbers for the SH modes. The thickness-dependent function $U_n(z)$ can be found in [46]. Similarly, the solutions for SH modes are given as:

$$\psi_{mn}^{>}(r,\theta) = H_m\left(l_n^h r\right)e^{im\theta}, \tag{10}$$

and

$$\psi_{mn}^{<}(r,\theta) = J_m\left(l_n^b r\right)e^{im\theta}. \tag{11}$$

As we have previously assumed a plane S0 wave incidence, the potential function for the incident field is given by:

$$\phi^{inc} = e^{ik_0^{Sh}x1} = e^{ik_0^{Sh}r\cos\theta} = \sum_{m=-\infty}^{\infty} i^m J_m\left(k_0^{Sh}r\right)e^{im\theta}. \tag{12}$$

According to Eq. (1), the displacement field of the incident wave can be further derived:

$$u_r^{inc} = \sum_{m=-\infty}^{\infty} i^m V_0^{Sh}(z)J_m'\left(k_0^{Sh}r\right)e^{im\theta}. \tag{13}$$

At a given frequency, Eq. (3) has numerous roots in the complex plane [15]. Retaining only the roots corresponding to forward propagating or evanescent modes, the scattering displacement field in the outer region can be expressed as:

$$\begin{aligned} u_r^{>} = \sum_{m=-\infty}^{\infty}\sum_{n=0}^{\infty}\Bigg\{ & a_{mn}V_n^{Sh}(z)H_m'\left(k_n^{Sh}r\right) + b_{mn}V_n^{Ah}(z)H_m'\left(k_n^{Ah}r\right)\\ & + imc_{mn}U_n^h(z)\frac{H_m\left(l_n^h r\right)}{l_n^h r}\Bigg\}e^{im\theta}. \end{aligned} \tag{14}$$

Similarly, the displacement in the inner region is:

$$\begin{aligned} u_r^{<} = \sum_{m=-\infty}^{\infty}\sum_{n=0}^{\infty}\Bigg\{ & d_{mn}V_n^{Sb}(z)J_m'\left(k_n^{Sb}r\right) + e_{mn}V_n^{Ab}(z)J_m'\left(k_n^{Ab}r\right)\\ & + imf_{mn}U_n^b(z)\frac{J_m\left(l_n^b r\right)}{l_n^b r}\Bigg\}e^{im\theta}. \end{aligned} \tag{15}$$

The expansion coefficients a_{mn} –f_{mn} need to be determined to obtain the scattered displacement fields. In addition to the radial displacement given above, other displacement components can also be derived through a similar procedure. Furthermore, by applying Hooke's law, the relevant stress components can also be determined. Next, we will determine the expansion coefficients by enforcing the boundary conditions of displacement and stress continuity at the defect, thereby

obtaining the solutions for the scattered field. For the displacement boundary condition, we have:

$$\boldsymbol{u}^{inc} + \boldsymbol{u}^{>} = \boldsymbol{u}^{<},\ 0 < z < 2b. \tag{16}$$

And for stress boundary condition:

$$\left(\boldsymbol{\sigma}^{inc} + \boldsymbol{\sigma}^{>}\right) \cdot \widehat{\boldsymbol{e}}_r = \boldsymbol{\sigma}^{<} \cdot \widehat{\boldsymbol{e}}_r, 0 < z < 2b, \tag{17}$$

where $\widehat{\boldsymbol{e}}_r$ is the unit vector in the radial direction. Substituting the displacement and stress components into the boundary conditions and solving for the coefficients yield the solutions for the scattered field.

2.1.2. Kirchhoff–Love plate theory

Through 3D theory, the scattered displacement fields in r, θ , and z directions can be obtained, corresponding to the scattered S0 wave, scattered SH wave, and scattered A0 wave, respectively. In this paper, we focus on the scattering of the S0 wave in z direction (out-of-plane displacement) after incidence. The out-of-plane displacement information is mainly provided by the scattered A0 mode. Therefore, the Kirchhoff–Love plate theory serves as an alternative approach for investigating the scattering problem. This model primarily captures the bending behavior of plates under small transverse loads, providing a foundation for analyzing the propagation of flexural waves within the structure [38]. The governing equation of the Kirchhoff–Love plate theory for an isotropic, homogeneous thin plate under free vibrations can be expressed as:

$$D\nabla^4 u(x,y,t) + \rho h \frac{\partial^2 u(x,y,t)}{\partial t^2} = 0, \tag{18}$$

where

$$D = \frac{E(2h)^3}{12(1-\nu^2)} \tag{19}$$

is the flexural stiffness of the plate,

$$\nabla^4 = \frac{\partial^4}{\partial x^4} + 2\frac{\partial^4}{\partial x^2 y^2} + \frac{\partial^4}{\partial y^4} \tag{20}$$

is the bi-harmonic operator and $u(x,y,t)$ represents the out-of-plane displacement at coordinates (x,y) at time t. After transformation, with $\zeta = \sqrt[4]{\frac{\rho h}{D}}$ as a variable, Eq. (18) can be expressed as:

$$\nabla^4 u(x,y,t) + \zeta \frac{\partial^2 u(x,y,t)}{\partial t^2} = 0. \tag{21}$$

Thus, the scattered A0 wave displacement can be obtained by solving Eq. (21).

2.2. Principle of physics-enhanced TransUNet (PTUNet)

The proposed TransUNet architecture, as schematically illustrated in Fig. 2(a), synergistically integrates the complementary strengths of UNet and Transformer models to address the multi-scale challenges in guided wave scattered wavefield reconstruction. UNet, as a classical encoder-decoder architecture, excels in image segmentation and feature extraction, making it well-suited for wavefield reconstruction [47]. Meanwhile, Transformer model, with its self-attention mechanism, effectively captures global dependencies, providing advantages in enhancing the representation capability of guided wave scattered fields [48]. This hybrid design synergistically integrates UNet's local pattern recognition capacity for wave scattering identification with Transformer's global dependency modeling for long-range wave propagation simulation.

TransUNet can obtain the scattered wavefield generated by defects with different geometric shapes and depth configurations. As depicted in Fig. 2(a), the network employs a symmetric encoder-decoder structure with four hierarchical processing stages. The encoder pathway performs progressive downsampling through sequential operations: dual 3 × 3 convolutional layers (LeakyReLU-activated), a 2 × 2 max-pooling layer, and a Transformer block incorporating multi-head attention mechanisms. These attention modules systematically capture long-range inter-layer dependencies across spatiotemporal coordinates, a critical enhancement over conventional convolutional approaches for wavefield prediction. The decoder executes mirrored upsampling via 2 × 2 transposed convolutions followed by feature concatenation and dual 3 × 3 convolutions, with skip connections bridging encoder-decoder stages to preserve high-resolution scattering features. The detailed description of the Transformer block and the entire PETUNet architecture can be found in Table 1 in Appendix. The architecture processes 101 × 101 thickness maps as inputs, producing 101 × 101 × 101 wavefield snapshots (400 mm × 400 mm spatial domain, $dx = dy = 4mm$) spanning 75 μs to 175 μs with 1 μs temporal resolution. The network's final output layer is configured with 101 channels, each representing one temporal snapshot. In this way, time is modeled as an additional "channel dimension," and the network jointly learns the spatiotemporal evolution in a single forward pass. The fundamentals of neural network architectures, including layer-wise forward propagation mechanisms, are well-documented in literatures and therefore omitted for brevity. For the thickness input configuration indexed by i, let $u_i(x,y,t)$ denote the reference wavefield obtained through FE simulations, while $u_i^*(x,y,t)$ represents the network prediction. The mean squared error (MSE) loss function governing the optimization process is formulated as:

$$\mathscr{L}_D = \frac{1}{N_D}\sum_{i=1}^{N_D}\left(u_i(x,y,t) - u_i^*(x,y,t)\right)^2, \tag{22}$$

where N_D corresponds to the spatiotemporal sampling points across the 101 × 101 × 101 domain. In this study, we compute $\mathscr{L}_D$ using all points within the domain. We refer to $\mathscr{L}_D$ as the data-driven loss since it solely represents the discrepancy between the predictions and the references. By providing labeled samples and iteratively minimizing $\mathscr{L}_D$, TransUNet learns the relationship between different defects and the corresponding scattered wavefields.

As previously discussed, a purely data-driven network heavily depends on the dataset and may produce predictions that violate physical laws. Therefore, unlike conventional PINNs which require no training data but are limited to a fixed input model, we enhance the original TransUNet by introducing Kirchhoff-Love equation constraints. This approach retains the core advantages of data-driven learning while incorporating physical constraints as corrections. The physics guidance ensures that the network not only fits the temporal outputs but also learns to preserve physical consistency across the entire time domain. Specifically, the network is initially trained using only data-driven constraints. Once the data-driven loss $\mathscr{L}_D$ ceases to decrease, the physical constraints are introduced, and the network undergoes a second round of training with both constraints applied simultaneously. In the early training stage, the data constraint is more effective in guiding the network to capture the main wavefield distribution, while the physics constraint provides long-term physical consistency. This strategy can obviously reduce some computational cost. The training process of PTUNet is illustrated in Fig. 2(b). Similar to the data-driven loss, the physics-driven loss $\mathscr{L}_P$ is expressed as:

$$\mathscr{L}_P = \frac{1}{N_P}\sum_{i=1}^{N_P}\left(\nabla^4 u_i^*(x,y,t) + \zeta\frac{\partial^2 u_i^*(x,y,t)}{\partial t^2}\right)^2, \tag{23}$$

where N_P is the spatiotemporal points randomly sampled from the computational domain. Partial derivative terms in the governing equation are obtained via the central FD scheme. The calculation process of the partial derivatives at $u_i^*(x,y,t)$ is described in Appendix. During the physics-constrained retraining phase, the composite loss function

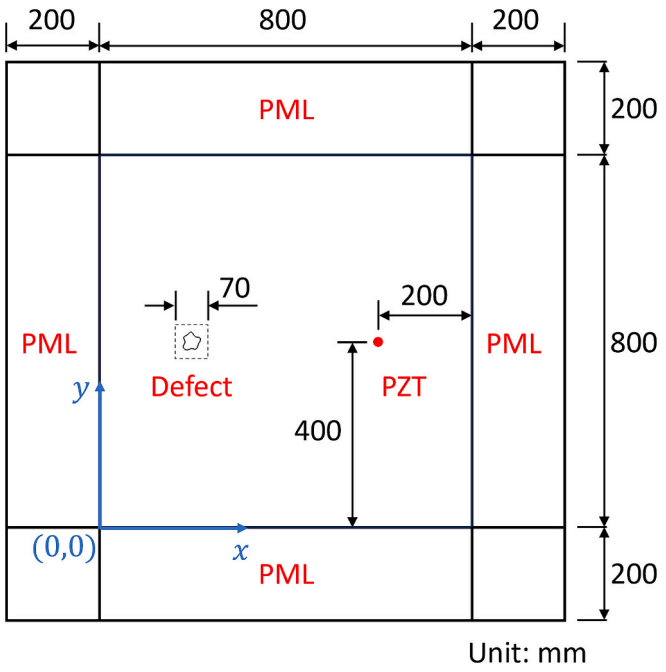


Fig. 3. Simulation Settings.

integrates both data-driven and physics- driven components as:

$$\mathscr{L} = \mathscr{L}_D + \lambda \mathscr{L}_P, \tag{24}$$

where λ denotes the adaptive weighting hyperparameter that balances the relative contributions of the data-driven and physics-driven constraints, ensuring numerical comparability between these two loss components. The selection of λ should be adjusted flexibly based on specific results, and the optimal value is determined through a process of trial and error.

3. Simulations and results

In this section, simulations were conducted on aluminum plates with irregularly shaped defects of varying depths to validate the enhancement effect of the physical constraints in the proposed method and to assess its effectiveness in solving the scattered wavefields.

3.1. Numerical simulation setup

We simulated the scattering of S0 Lamb waves by irregular defects in an aluminum plate using FE software. As shown in Fig. 3, the aluminum plate measures 800 mm in length, 800 mm in width, and 1.5 mm in thickness, with a density of $\rho = 2700$ kg/m^3, Young's modulus $E = 70.7$ GPa, and Poisson's ratio $\nu = 0.33$. The defects, characterized by arbitrary irregular shapes and varying depths, are centered at (200, 400) and constrained within a 70 mm $\times$ 70 mm square. For each specific defect, the thickness remains constant. A symmetric dual-sensor excitation method was employed to generate S0 Lamb wave. A piezoelectric transducer with a center frequency of 100 kHz was positioned at (600, 400), while another identical transducer was attached to the opposite side of the plate. To mitigate boundary reflections, 200 mm-wide perfectly matched layers (PML) were applied around the plate. In FE simulation, the excitation signal was a five-cycle Hanning-windowed sinusoidal function with a center frequency of 100 kHz. The time step was set to 1×10^{-7} s, with a total simulation duration of 2.5×10^{-4} s. In this study, a maximum mesh size of 5 mm was adopted, with local refinement applied near the defect. This resolution, together with local refinement, allowed both modes to be reasonably resolved while keeping the computational cost manageable.

Fig. 4 presents the FE simulation results of a blind hole and their comparisons with 3D theory. The blind hole is located at (200, 400), with a radius of 25 mm, and a remaining thickness of 0.75 mm. The simulation provides 3D scattered wavefield data. Fig. 4(a) shows an out-of-plane displacement snapshot from FE simulation at 150 μs. As observed in Fig. 4(a), the incident S0 wave has propagated to the left boundary of the plate, generating scattered waves (S0-A0 waves) upon interacting with the defect. Fig. 4(b) demonstrates that the far-field amplitudes of out-of-plane displacement obtained from FE simulations aligns well with the theoretical predictions of 3D theory. This agreement validates the accuracy of FE parameter settings and confirms the reliability of our simulations in capturing the scattering behavior of S0 Lamb waves.

We extracted the out-of-plane displacement within a square region spanning $x = 0$ to 400 mm and $y = 200$ to 600 mm over a time window of 75 μs to 175 μs. Since the S0 Lamb wave reaches the defect region at approximately 75 μs, the preceding wavefield does not contain any scattering information. To better focus on the relevant data for solving the scattering problem, we apply spatial and temporal truncation, ensuring that the extracted wavefield primarily captures the useful scattering characteristics. The resulting wavefield dataset had dimensions of 101 $\times$ 101 $\times$ 101 in x, y, and t. A total of 400 random defect

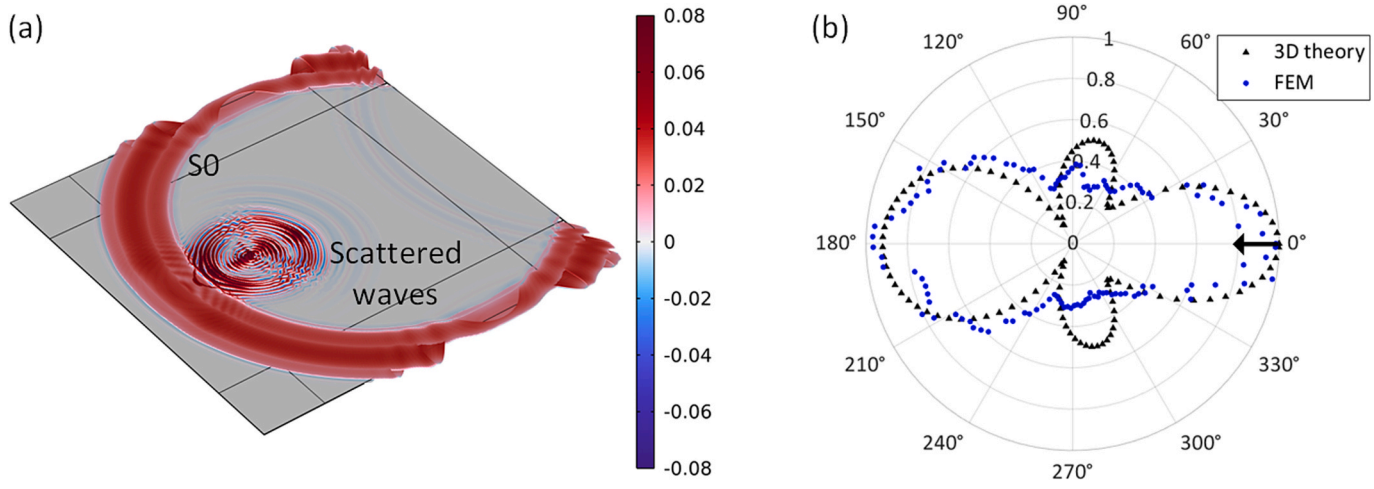


Fig. 4. The scattering results of blind holes. (a) A snapshot of FE simulation. (b) Comparison of far-field amplitudes of scattered waves (black arrow represents incident direction).

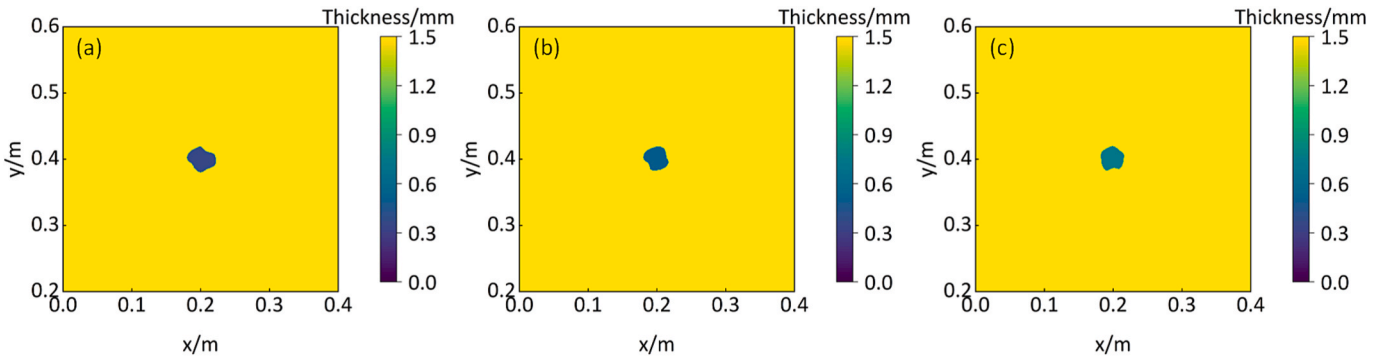


Fig. 5. Thickness distribution maps of three defects in the testing set.

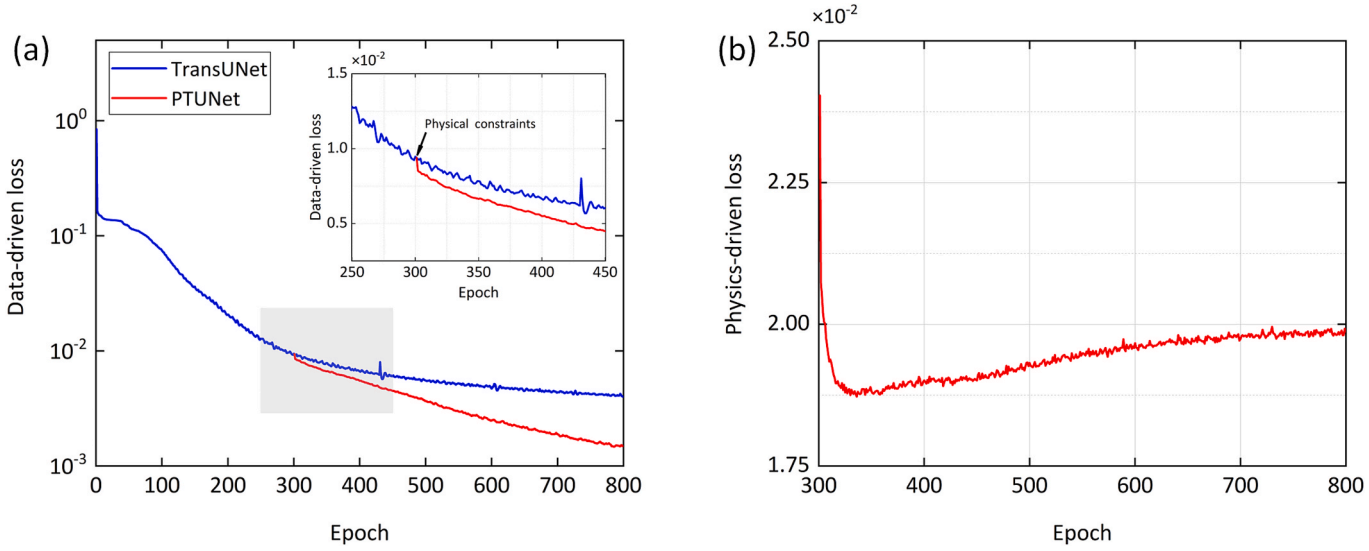


Fig. 6. Training loss curves. (a)Data-driven loss. (b) Physics-driven loss.

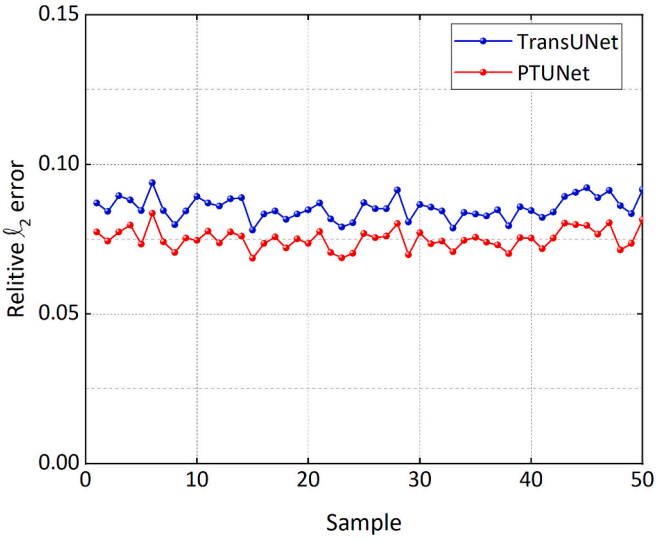


Fig. 7. The comparison of relative error on testing set.

scattering simulations were performed to generate the dataset for network training, with 350 samples used for training and the remaining 50 samples for testing. The remaining thickness of the defect ranges from 0.3 mm to 1 mm. Fig. 5 presents three different defects from the testing set, with increasing remaining thicknesses of 0.3655 mm, 0.5000 mm, and 0.7322 mm, respectively. In the following sections, we use these three defect cases to evaluate the capability of the proposed method in solving the scattered wavefields.

3.2. *Verification of the enhancement effect of physical constraints*

To evaluate the enhancement effect of physical constraints on solving the scattered wavefield, we compared the training process and test results of TransUNet, which relies solely on data constraints, with PTUNet, which incorporates both data and physical constraints. Fig. 6 illustrates the training loss curves for both methods. The learning rate is 0.0001, and the batch size is set to 1. During the first 300 iterations, training is conducted using only data constraints. As the data-driven loss begins to plateau at the 300th iteration, physical constraint is introduced with 2000 random collocation points for each time snapshot (N_P = 2000, 20 % of the total points) and the weight of 0.01 ($\lambda = 0.01$). We determine the optimal weight values through grid searches over $\lambda = 0.1$, 0.01, and 0.001. As shown in Fig. 6(a), the embedding of physical constraints effectively accelerates the training process and reduces the overall training loss. Fig. 6(b) demonstrates that the physics-driven loss also decreases with iterations and eventually stabilizes after embedding the physical constraint.

To further evaluate the generalization capability of PTUNet, we compared the relative ℓ_2 error of the two methods on the testing set. The relative ℓ_2 error is computed as follows:

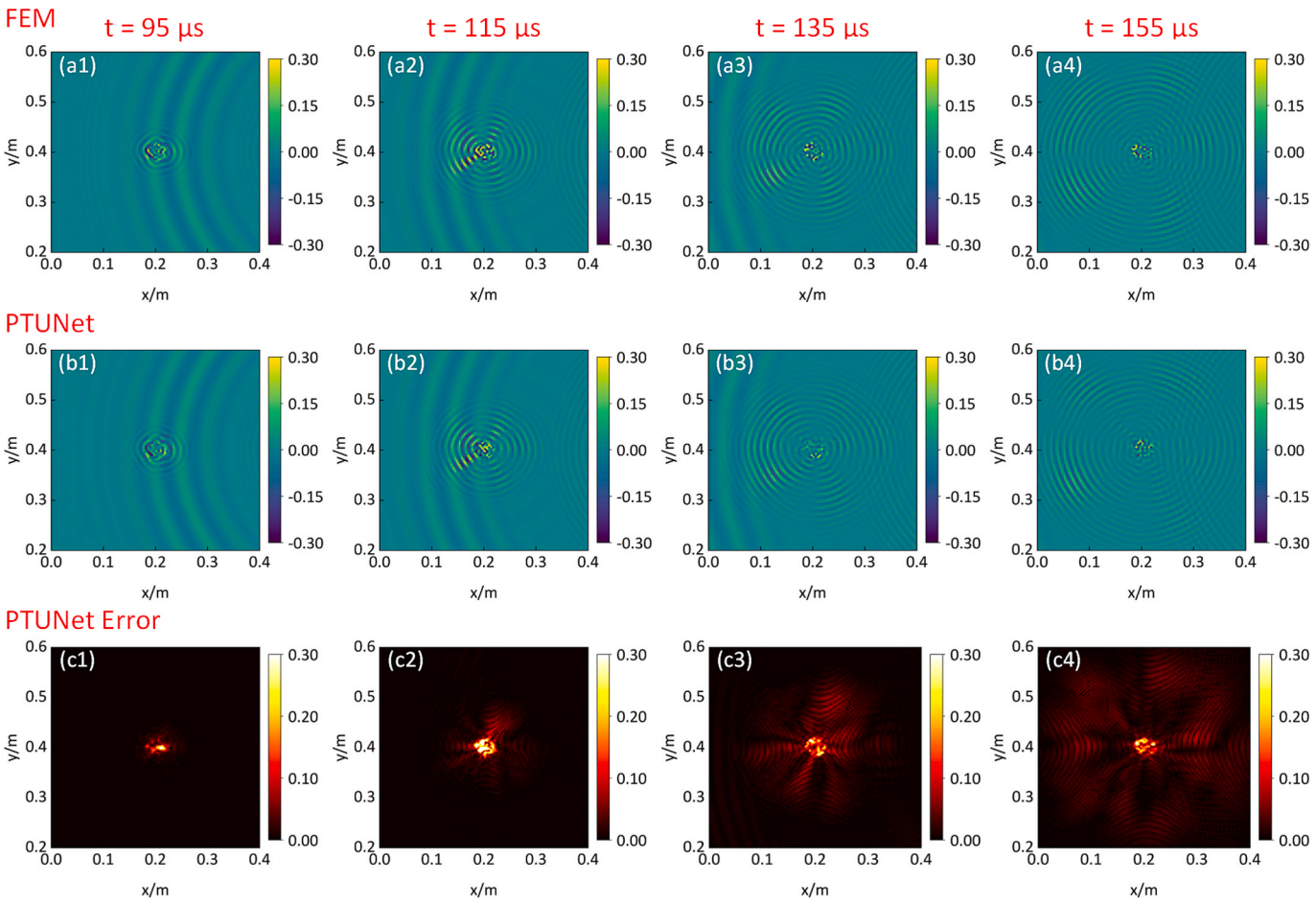


Fig. 8. The comparisons of the scattered wavefields of the defect shown in Fig. 5(a).

$$\ell_2 = \frac{\left\| u_i(x,y,t) - u_i^*(x,y,t) \right\|_2}{\left\| u_i(x,y,t) \right\|_2}. \tag{25}$$

For each sample in the testing set, there are 101 time-domain snapshots of the scattered wavefield. We compute the ℓ_2 error for each sample by averaging its relative ℓ_2 errors across all time steps, as shown in Fig. 7. The results indicate that PTUNet consistently achieves lower overall error compared to TransUNet, demonstrating superior performance on the testing data and enhanced generalization ability. This further validates the effectiveness of incorporating physical constraints, which not only guide the model to adhere to physical laws and accelerate convergence but also reduce dependency on training data. Consequently, for solving wave propagation problems and other physics-informed tasks, deep learning models with integrated physical constraints can effectively improve both accuracy and generalization.

3.3. Simulation results of PTUNet

3.3.1. Comparisons of wavefield

After validating the effectiveness of incorporating physical constraints, this section focuses on solving the S0 Lamb wave scattering process using PTUNet. Figs. 8 to 10 correspond to the scattered wavefields of the three randomly generated defects shown in Fig. 5(a), (b), and (c), respectively. In Fig. 8, (a1) to (a4) present the out-of-plane displacements obtained from 3D FE simulations, serving as references for the scattered wavefields. Fig. 8(b1) to (b4) and (c1) to (c4) illustrate the PTUNet predictions and their respective error distributions relative to FE solutions. To comprehensively capture the wave propagation and scattering dynamics, we examine the wavefields at four distinct time instants—95 μs, 115 μs, 135 μs, and 155 μs. As shown in Fig. 8(a1), at 95 μs, S0 Lamb wave reaches the defect, initiating the formation of scattered waves (S0-A0). By 115 μs, S0 wave continues propagating, while the scattered waves radiate outward from the defect. At this stage, another wave emerges on the right side of the computational domain, corresponding to the A0 mode generated during excitation. At 135 μs, as depicted in Fig. 8(a3), S0 Lamb wave is on the verge of exiting the domain, while the scattered waves have begun interacting with A0 mode, resulting in wave interference. By 155 μs, the originally excited S0 Lamb wave has completely exited the domain, leaving behind only the scattered waves and the approaching A0 mode, which will interact with the defect and generate secondary scattered waves (A0-A0), further increasing the complexity of the wavefield. It is noteworthy that as wave propagation progresses, the amplitude of the excited S0 wave gradually attenuates. In contrast, the amplitude of the scattered waves initially increases as they form and then gradually diminishes over time. Additionally, the wavefields exhibit anisotropic amplitude variations across different propagation angles, highlighting the influence of defect asymmetry on the scattering characteristics. This intrinsic correlation between defect geometry and the scattered wavefield underscores the feasibility of inferring defect properties through wave scattering analysis.

The comparisons between FE simulations and predictions indicate that PTUNet effectively captures the scattered wavefield throughout wave propagation. The wave propagation obtained from PTUNet closely align with those from FE simulations, demonstrating that the proposed method has successfully learned the scattering characteristics. As observed in the error distributions from Figs. 8 to 10, discrepancies are primarily concentrated around the defect regions and theirs surrounding

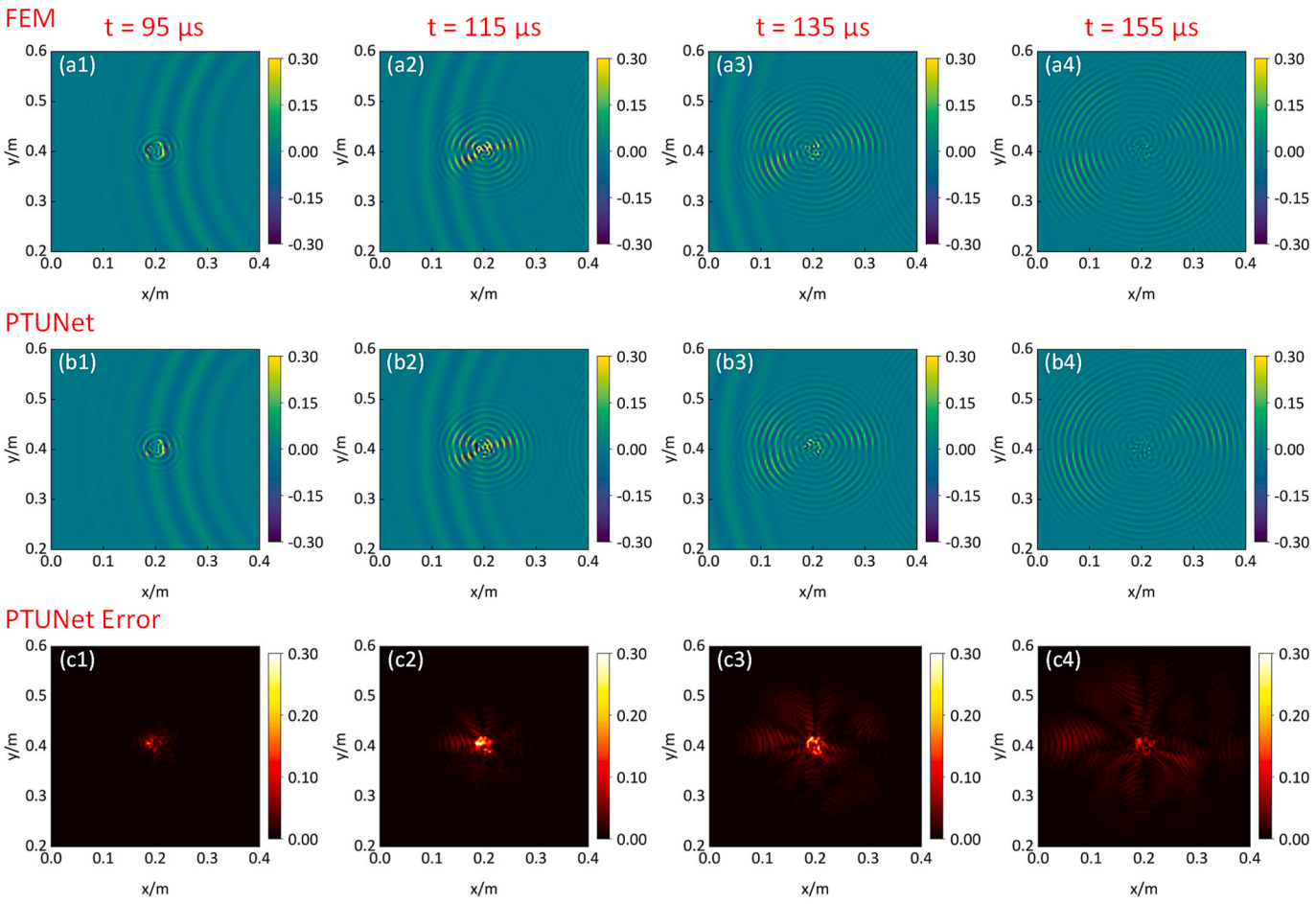


Fig. 9. The comparisons of the scattered wavefields of the defect shown in Fig. 5(b).

wave interference zones. Additionally, as wave propagation, the errors tend to increase in the far-field region. This may be attributed to PTUNet's reduced prediction accuracy at longer time steps during training or the inherently complex nature of wave scattering induced by the defect. PTUNet needs to capture intricate physical details, and its predictive capability faces challenges in fully reconstructing all fine-scale features. Despite these local errors, the results presented in Figs. 8 to 10 confirm that PTUNet can solve the scattered wavefields of defects with different shapes and thicknesses. The model exhibits reasonable generalization ability and maintains high physical consistency, making it a promising approach for wave scattering analysis in complex structures.

3.3.2. *Comparisons of signals and far-field amplitudes of the scattered waves*

After obtaining the scattered wavefield for irregular defects, the time-domain signal at any arbitrary location within the solution domain can be extracted. To systematically analyze the scattering characteristics, we define the coordinate (200, 400), the center of the domain, as the origin and uniformly distribute 360 receiver points along a circular trajectory with a radius of 160 mm. This configuration enables the acquisition of time-domain signals at each receiver location, facilitating a detailed examination of the wave propagation and scattering behavior. Fig. 11 presents a comparative analysis of the scattered signals corresponding to the three defects shown in Fig. 5(a), (b), and (c). The signals are sampled at 12 angular positions with 30° intervals, providing a representative overview of the wave propagation characteristics. As the observation angle increases, the propagation distance of S0 mode initially increases and subsequently decreases symmetrically about 180°, resulting in an arc-like distribution of the signals centered at 180°. When S0 wave reaches the defect, scattered A0 waves (S0-A0) are generated and propagate outward around the defect. Since the propagation paths to the receiver locations are nearly identical, the scattered A0 waves are expected to arrive at approximately the same time across different angular positions. The results in Fig. 11 further corroborate this observation, further validating the physical consistency of the scattering process.

From a global perspective, the signals predicted by PTUNet exhibit a good agreement with FE simulations. In the S0 wave region where scattering has not yet occurred, PTUNet's predictions align almost perfectly with FE, demonstrating the model's ability to accurately capture the undisturbed wave propagation. After approximately 135 μs, when the scattered A0 wave reaches the receiver locations, slight discrepancies in amplitude emerge between predictions and FE results. Nevertheless, the overall wave propagation trend remains consistent across both methods. These minor deviations may be attributed to PTUNet's inherent limitations in learning fine-scale scattering features, which could lead to reduced accuracy in capturing wave interactions.

To further examine the scattering characteristics, we extracted the peak amplitude of the scattered A0 waves at different angular positions, yielding the results of the far-field amplitudes of the scattered waves presented in Fig. 12. The results illustrate that the predicted scattering energy distribution closely aligns with FE simulations, exhibiting a similar scattering pattern. This further substantiates PTUNet's ability to generalize across different scatterers. Notably, in regions where the scattering features exhibit pronounced variations, such as near 90° and 270° in Fig. 12(b), PTUNet successfully captures these variations with good fidelity. While some discrepancies are observed in specific angular ranges, such as between 150° and 180° in Fig. 12(a) and (c), the overall

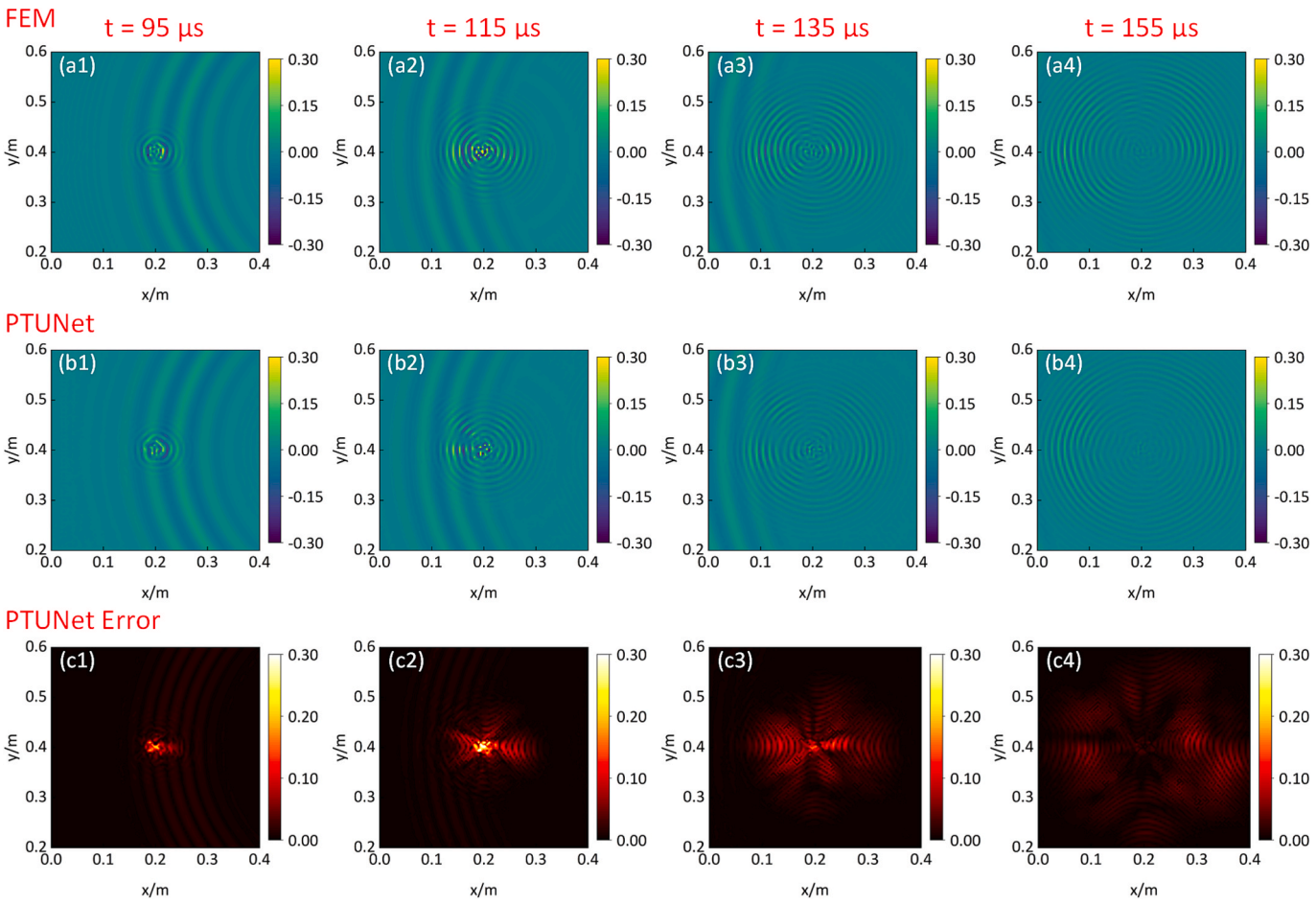


Fig. 10. The comparisons of the scattered wavefields of the defect shown in Fig. 5(c).

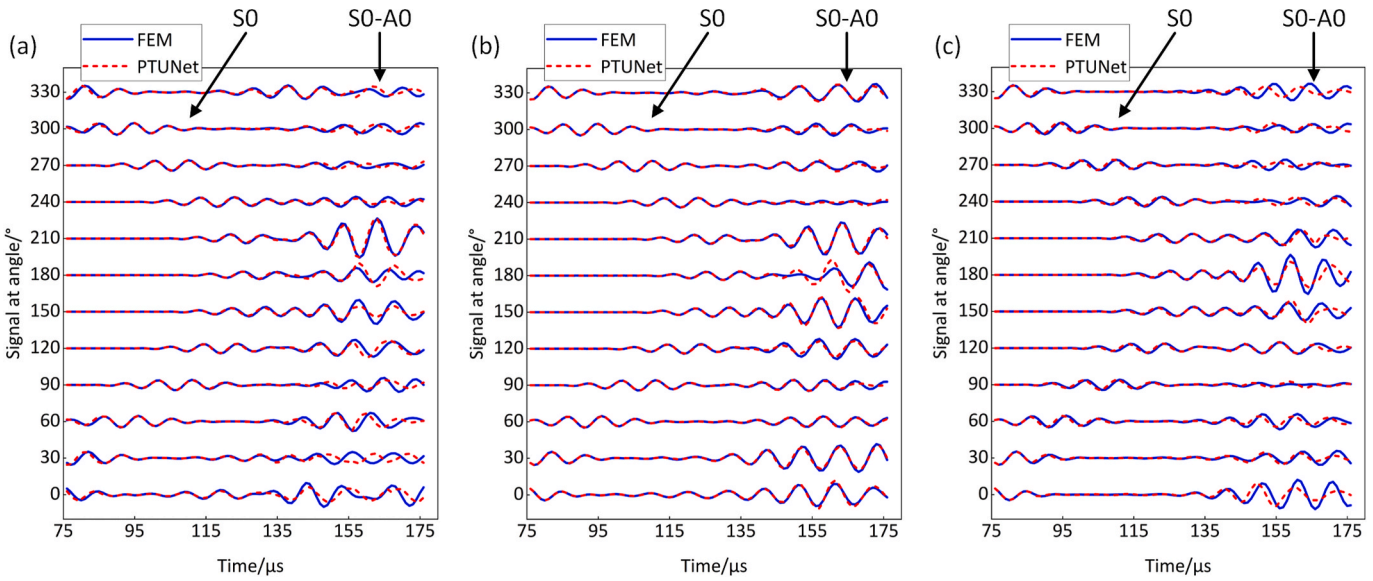


Fig. 11. The comparisons of the signals of the three defects shown in Fig. 5.

trend remains well-aligned with FE simulations. These deviations may stem from several factors such as the high complexity of certain scattering behaviors. Overall, PTUNet demonstrates a considerable capacity for predicting the propagation characteristics of scattered waves. Future efforts could focus on refining the physical constraint mechanisms to further enhance the predictive performance and generalization of the model.

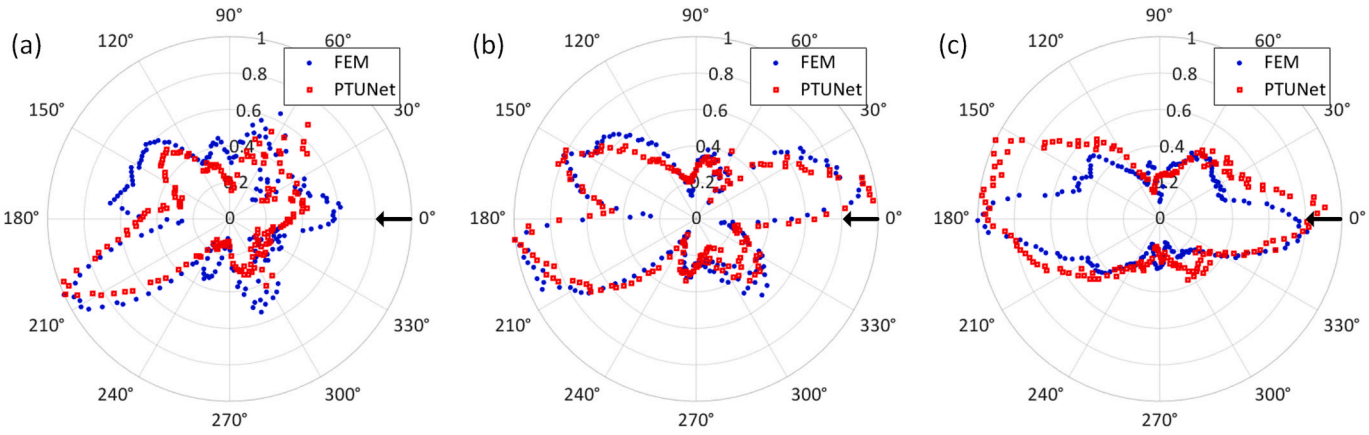


Fig. 12. The comparisons of the far-field amplitudes of the scattered waves of three defects shown in Fig. 5 (black arrow represents incident direction).

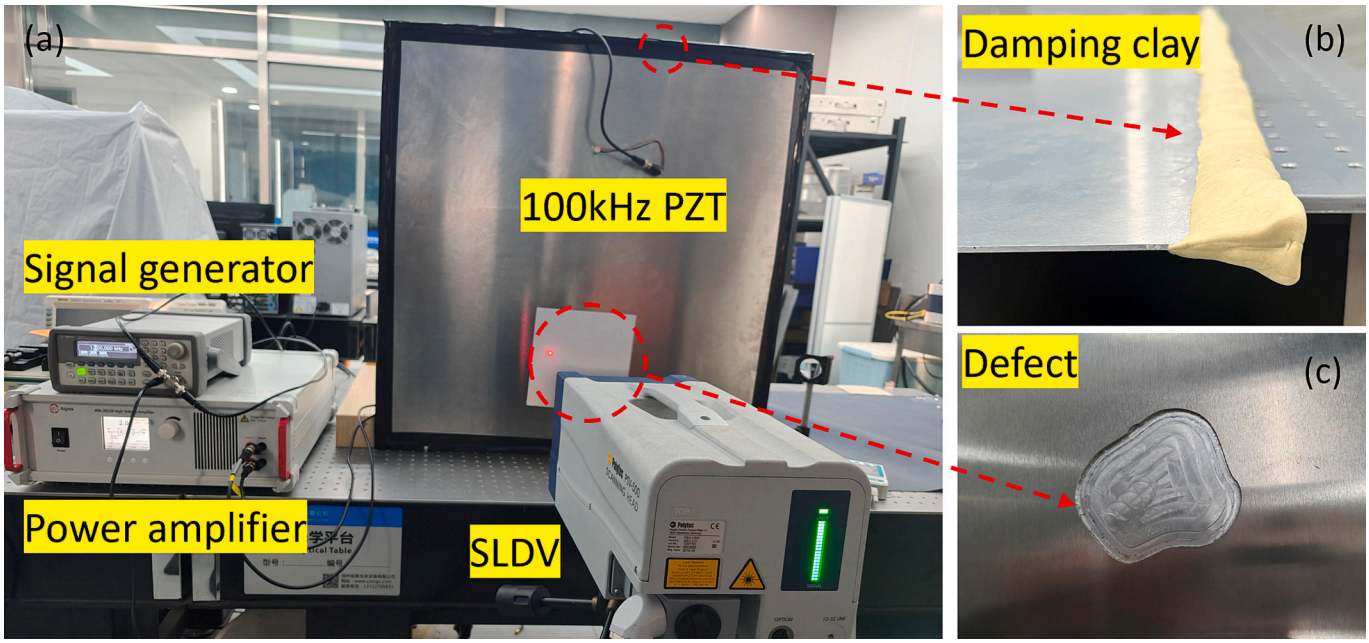


Fig. 13. Setup of the SLDV acoustic field scanning experiment.

4. Experiments and results

4.1. Experiment setup

To experimentally validate the performance of the proposed method in solving the scattered wavefield, we conducted a field scanning experiment using a Polytec SLDV. The experimental setup is shown in Fig. 13. A 6061 aluminum plate with dimensions of 800 × 800 × 1.5 mm was securely mounted on an optical platform. To minimize boundary reflections, damping clay was applied along the plate's edges. Specifically, the clay was shaped into a wedge, gradually tapering in thickness from the plate's edge toward its center, as illustrated in Fig. 13(b). This design effectively reduces secondary reflections caused by wave interactions with the damping material. An irregularly shaped defect with a residual thickness of 0.5 mm was precisely machined onto the aluminum plate using a milling process, as shown in Fig. 13(c). The spatial coordinates of the defect were kept consistent with those used in FE simulations, ensuring a rigorous evaluation of the predictive accuracy of the proposed method. Two identical 100 kHz piezoelectric transducers were symmetrically mounted on the aluminum plate at a distance of 400 mm from the defect center to generate a pure S0 Lamb wave. A five-cycle sinusoidal excitation signal with a frequency of 100 kHz and an amplitude of 4 V peak to peak was generated by an arbitrary waveform generator. The signal was then amplified by a factor of 20 using an Aigtek ATA-2022B dual-channel power amplifier before being applied to the two transducers. The SLDV probe was carefully adjusted to ensure that the emitted laser beam was as perpendicular as possible to the plate surface. To enhance the probe's ability to capture vibrations, a reflective material was applied to the defect surface. The scanning region was defined as a 100 mm × 100 mm square centered on the defect. The SLDV operated at a sampling frequency of 6.25 MHz, with a total of 5000 acquisition points. A band-pass filter was applied in the range of 80 kHz to 120 kHz to isolate the relevant frequency components.

4.2. Comparisons of scattering results

We input the thickness distribution of the defect into the pre-trained PTUNet model to generate the predicted scattered wavefields and compare them with FE results. As shown in Fig. 14, PTUNet accurately captures the propagation of S0 waves and the scattering behaviors after interacting with the defect, showing good consistency with FE simulations.

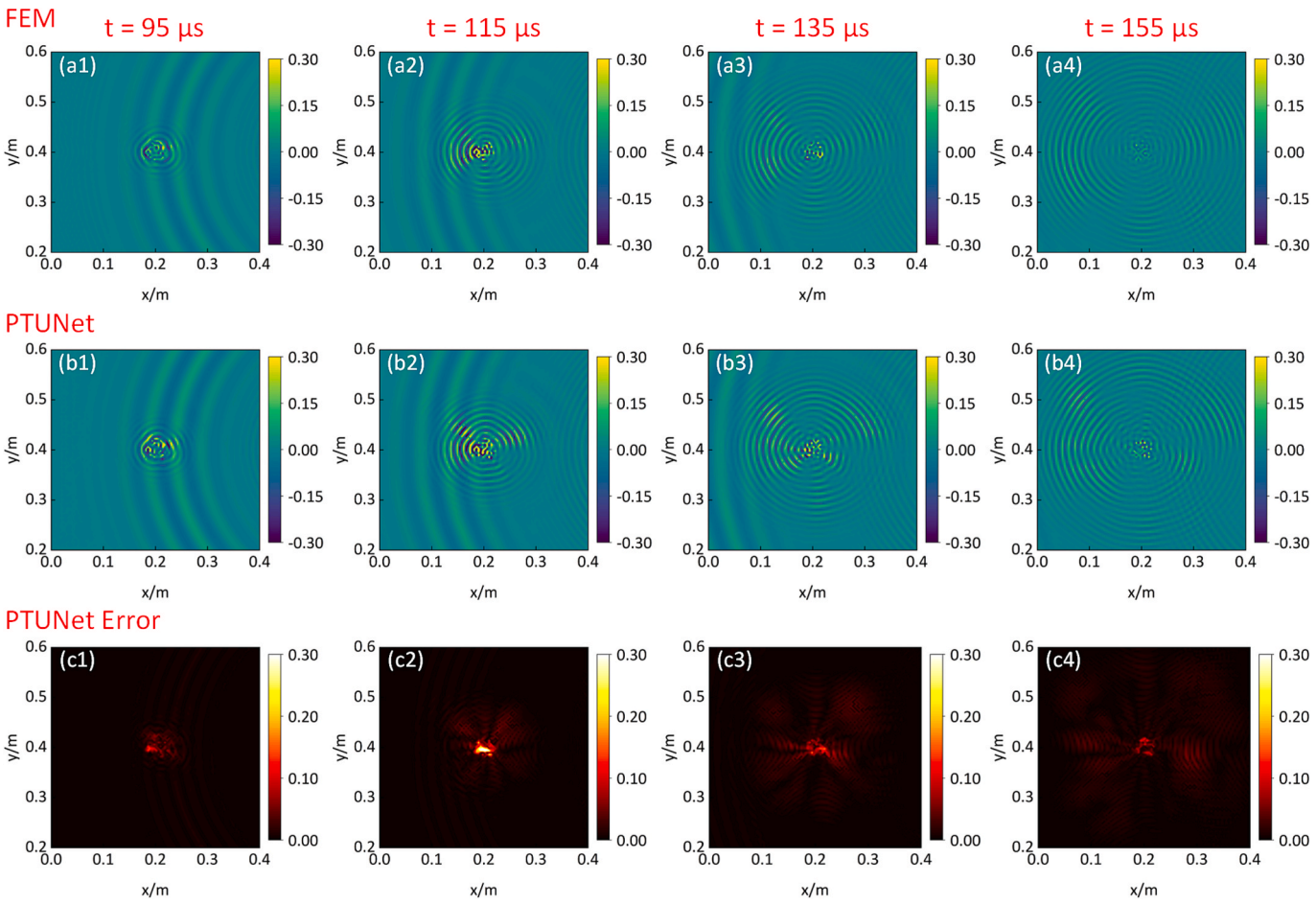


Fig. 14. The comparisons of the scattered wavefields of the defect in experiments.

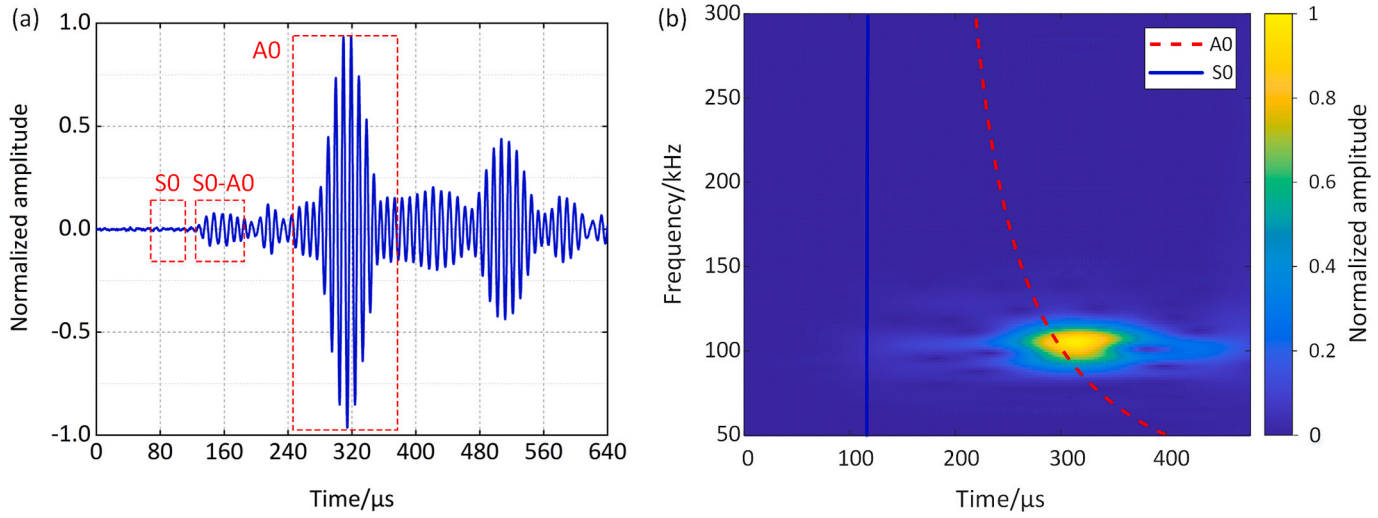


Fig. 15. Analysis of the experimental signal. (a) Time domain signal at the receiving point of 180°. (b) Continuous wavelet transform analysis.

Through SLDV scanning, we obtained the scattered wavefields of the defect in experiment. Assuming 360 receivers are uniformly distributed along a circular path with a radius of 100 mm centered on the defect, the time-domain signal and its continuous wavelet transform (CWT) at the 180° receiving point are presented in Fig. 15. Given the wave structure of S0 mode, its out-of-plane displacement is relatively small, making it challenging to capture with SLDV. Consequently, as shown in Fig. 15(a), the amplitude of S0 mode is significantly small. Based on the propagation distance, S0 wave is expected to appear around 90 μs, while the scattered wave generated by the defect emerges near 135 μs. The scattered wave is more pronounced because, compared to S0 mode, A0 mode exhibits a larger out-of-plane displacement. The most dominant wave component, A0 mode generated by the excitation, appears around 240 μs with the highest amplitude. The CWT of the signal further reveals

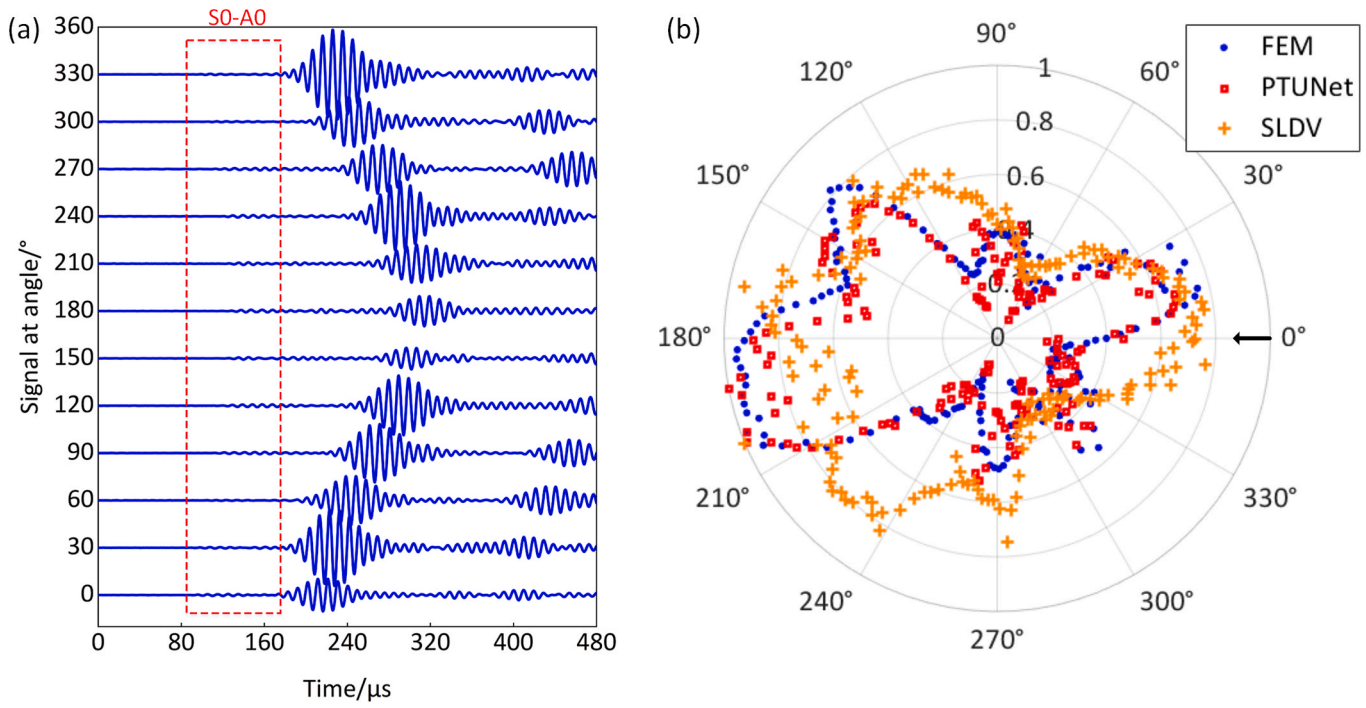


Fig. 16. (a) Experimental time-domain signals. (b) Comparison of far-field amplitude of scattered waves (black arrow represents incident direction).

Table 1
Network architecture & hyperparameters.

Module	Configuration	Output size
Input	Thickness map (1 channel)	1 × 101 × 101
Encoder-1	Conv2d (1 → 16), LeakyReLU(0.01) Conv2d (16 → 16), LeakyReLU(0.01)	16 × 101 × 101
Transformer-1	MultiheadAttention (embed = 16, heads = 4), FFN (16 → 32 → 16), LayerNorm	16 × 101 × 101
Encoder-2	Conv2d (16 → 32), LeakyReLU(0.01) Conv2d (32 → 32), LeakyReLU(0.01)	32 × 50 × 50
Transformer-2	MultiheadAttention (embed = 32, heads = 4), FFN (32 → 32 → 32), LayerNorm	32 × 50 × 50
Encoder-3	Conv2d (32 → 64), LeakyReLU(0.01) Conv2d (64 → 64), LeakyReLU(0.01)	64 × 25 × 25
Transformer-3	MultiheadAttention (embed = 64, heads = 4), FFN (64 → 32 → 64), LayerNorm	64 × 25 × 25
Encoder-4	Conv2d (64 → 128), LeakyReLU(0.01) Conv2d (128 → 128), LeakyReLU(0.01)	128 × 13 × 13
Transformer-4	MultiheadAttention (embed = 128, heads = 4), FFN (128 → 32 → 128), LayerNorm	128 × 13 × 13
Bottleneck	Conv2d (128 → 256), LeakyReLU(0.01) Conv2d (256 → 256), LeakyReLU(0.01)	256 × 7 × 7
Decoder-1	ConvTranspose2d (256 → 128) + skip Conv2d (256 → 128), LeakyReLU(0.01) Conv2d (128 → 128), LeakyReLU(0.01)	128 × 14 × 14
Decoder-2	ConvTranspose2d (128 → 64) + skip Conv2d (128 → 64), LeakyReLU(0.01) Conv2d (64 → 64), LeakyReLU(0.01)	64 × 28 × 28
Decoder-3	ConvTranspose2d (64 → 32) + skip Conv2d (64 → 32), LeakyReLU(0.01) Conv2d (32 → 32), LeakyReLU(0.01)	32 × 56 × 56
Decoder-4	ConvTranspose2d (32 → 16) + skip Conv2d (32 → 16), LeakyReLU(0.01) Conv2d (16 → 16), LeakyReLU(0.01)	16 × 112 × 112
Output	Conv2d (16 → 101), bilinear resize to 101 × 101	101 × 101 × 101

its time–frequency characteristics, as shown in Fig. 15(b). The frequency is mainly distributed around 100 kHz. Due to the dominant amplitude of the excitation-induced A0 mode, the maximum energy is observed around 300 μs, aligning well with theoretical expectations.

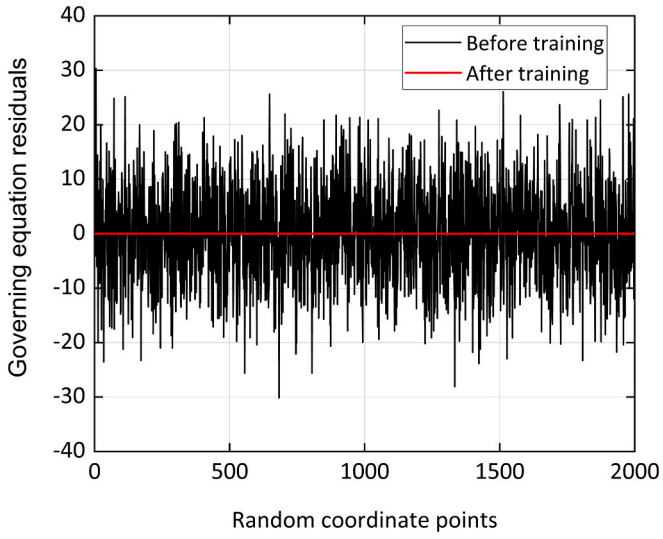


Fig. 17. Comparison of governing equation residuals before (black) and after (red) training.

Fig. 16(a) presents the time-domain signals obtained from 12 receivers, which uniformly distributed at 30° intervals. These signals are utilized to derive the far-field amplitudes of the scattered waves obtained from SLDV scanning. The comparisons of FE simulation, PTUNet's prediction, and SLDV scanning are shown in Fig. 16(b). Despite the relatively smooth and simple geometry of the defect boundary, the scattered wavefield exhibits significant complexity, as evidenced by obvious amplitude variations in the scattering result. The PTUNet predictions demonstrate good agreement with FE simulations, effectively and comprehensively capturing the intricate scattering characteristics. The proposed method successfully reconstructs the majority of the scattering information, underscoring its capability to model wave interactions with irregular defects accurately. Although discrepancies exist between the experimental scanning and FE simulations, the overall

trend remains consistent. Considering factors such as the inherently weak out-of-plane scattering amplitude, unavoidable boundary reflections, machining tolerances of the defect, and various experimental noise sources, it is reasonable to conclude that the SLDV experimental results are in qualitative agreement with both FE simulations and PTUNet predictions, reinforcing their mutual validation.

5. Discussion

5.1. Advantages of the proposed method

By integrating the local feature extraction capabilities of UNet with the global modeling power of Transformer, and further constraining the solution with Kirchhoff-Love plate theory, PTUNet demonstrates outstanding performance in solving S0 Lamb wave scattering induced by irregular defects in plate structures. Notably, the irregular defect size is approximately one-tenth of the wavelength of the excitation signal, leading to highly intricate scattering characteristics, including mode conversion and other fine wavefield details. Through the enhancement of physical constraints, PTUNet achieves accurate predictions of subtle scattering features with a limited training dataset, yielding a lower mean relative ℓ_2 error on the testing set compared to purely data-driven training approach. Unlike BEM or hybrid numerical methods, PTUNet eliminates the need for repeated modeling of different configurations and does not require mesh discretization, thereby avoiding the substantial computational costs associated with high-frequency simulations. In computing the partial derivatives required for physical constraints, we adopt FD methods instead of automatic differentiation. This choice is driven by the fact that employing automatic differentiation would necessitate including spatial coordinates x, y and temporal coordinate t as additional network inputs. Consequently, the input dimensionality would increase from 2D thickness representations to a 4D dataset incorporating spatial and temporal information, significantly escalating computational complexity.

It is crucial to highlight that the proposed method offers valuable insights that can inspire solutions for a broader range of problems. A particularly promising application lies in defect inversion based on scattering information. By leveraging the enhancement of physical constraints, it becomes possible to reconstruct the thickness and shape of defects approximately one-tenth of the incident wavelength, representing an exciting research avenue. Furthermore, the inherent generalizability and scalability of this approach pave the way for its extension to other fields, as the underlying physical model can always be associated with appropriate governing equations. This versatility underscores the method's potential as a powerful tool for advancing interdisciplinary research and fostering new developments in scientific and engineering applications.

5.2. Limitations and future works

In this study, the defect models remain relatively simplified, as the residual thickness within each defect is assumed to be uniform. This assumption differs significantly from real-world defects, which always exhibit complex and irregular thickness variations. The primary reason for this simplification is the limited thickness (1.5 mm) of the tested plate, which inherently constrains the range of residual thickness variations. Minor variations in thickness are unlikely to induce substantial changes in scattering characteristics. Furthermore, due to the considerable computational cost of 3D FE simulations, it is time-consuming to generate a much larger dataset for training. Capturing the subtle variations in the scattered wavefield caused by small thickness changes remains a formidable challenge. To address these limitations, we propose two key research directions. First, we plan to explore the potential of neural operators for solving scattered wavefields, aiming to enhance feature extraction and self-learning capabilities while reducing the network complexity. Second, we plan to incorporate additional and more precise governing equations to better describe wave propagation, thereby improving the model's ability to capture intricate scattering behaviors with higher fidelity.

Furthermore, our experimental results are not yet perfect. This is primarily due to challenges in exciting a pure S0 mode during the experiment. Several factors may contribute to the discrepancies in the results, including the asymmetric arrangement of the excitation transducers, differences in material properties of the aluminum plate compared to the simulations, and unavoidable boundary reflections. Additionally, the out-of-plane displacement of S0 mode is relatively small compared to A0 mode, which means that the scattered wave information may be submerged in noise, making it difficult to capture with high precision.

Finally, this study has primarily focused on solving the out-of-plane displacement of the scattered wavefield. Moving forward, we intend to explore more general wave equations to address the solution of 3D scattering characteristics. Additionally, we will investigate alternative implementations of physical constraints, such as using integral form constraints or leveraging Chebyshev spectral methods to accelerate the calculation of higher-order derivatives. These advancements will enhance the accuracy and generalization of the model in solving complex scattering problems.

6. Summary

In this study, we propose a physics-enhanced TransUNet (PTUNet) framework to solve the scattered wavefield of irregular defects in plate structures. This method possesses the capability of extracting local features and modeling global characteristics of the scattered wavefield. Moreover, with the physical constraints imposed by Kirchhoff-Love plate theory, it enables accurate capture of fine-scale features in complex scattered wavefields. We conducted simulations on three sets of randomly generated defects and compared the results with FE simulations. The proposed method effectively predicts the propagation of S0 Lamb waves and their mode conversion upon interacting with defects, capturing the intricate scattering characteristics. The predicted time-domain signals and the far-field amplitudes of scattered waves from PTUNet align well with FE results. Additionally, we performed acoustic field scanning experiments using SLDV to measure the out-of-plane displacement of scattered waves from a random defect on a 1.5 mm-thick aluminum plate. The scattering result predicted by PTUNet exhibits a reasonable agreement with both FE simulation and experimental measurement. These simulation and experimental results validate the effectiveness of the proposed method in solving S0 Lamb wave scattering problems. Future work includes exploring neural operators to improve predictive performance, investigating alternative governing equations as physical constraints to achieve more accurate 3D wavefield solutions, and studying different implementations of physical constraints for further improvements.

CRediT authorship contribution statement

Linfeng Wang: Writing – original draft, Visualization, Methodology. **Hongyan Zhang:** Visualization, Software, Methodology. **Zhen Zhang:** Validation, Methodology. **He Sun:** Software, Formal analysis. **Xiao Ying:** Validation, Software. **Jing Hou:** Investigation, Funding acquisition. **Jian Li:** Writing – review & editing, Supervision, Methodology. **Zhoumo Zeng:** Writing – review & editing, Supervision, Funding acquisition. **Yang Liu:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Science Foundation of China under Contract No. 12474459, in part by the Natural Science Foundation of Zhejiang Province under Contract No. LY23E010001.

Appendix A

The detailed description of PETUNet architecture is presented in Table 1.

The partial derivatives at $u_i^*(x,y,t)$ using the central finite difference scheme are described as

$$\frac{\partial^4 u_i^*}{\partial x^4} = \frac{u_i^*(x+2,y,t) - 4u_i^*(x+1,y,t) + 6u_i^*(x,y,t) - 4u_i^*(x-1,y,t) + u_i^*(x-2,y,t)}{(\Delta x)^4}, \tag{A.1}$$

$$\frac{\partial^4 u_i^*}{\partial y^4} = \frac{u_i^*(x,y+2,t) - 4u_i^*(x,y+1,t) + 6u_i^*(x,y,t) - 4u_i^*(x,y-1,t) + u_i^*(x,y-2,t)}{(\Delta y)^4}, \tag{A.2}$$

$$\begin{aligned}\frac{\partial^4 u_i^*}{\partial x^2 \partial y^2} = \frac{1}{(\Delta x)^2(\Delta y)^2}[&u_i^*(x+1,y+1,t) + u_i^*(x+1,y-1,t) + u_i^*(x-1,y+1,t) \\ &+ u_i^*(x-1,y-1,t) - 2u_i^*(x-1,y,t) - 2u_i^*(x,y+1,t) - 2u_i^*(x,y-1,t) \\ &+ 4u_i^*(x,y,t)],\end{aligned} \tag{A.3}$$

$$\frac{\partial^2 u_i^*}{\partial t^2} = \frac{u_i^*(x,y,t+1) - 2u_i^*(x,y,t) + u_i^*(x,y,t-1)}{(\Delta t)^2}. \tag{A.4}$$

To validate the effectiveness of FD approximations in the calculation of physics-driven loss, we compare the pre-training and post-training governing equation residuals at 2000 randomly sampled coordinates for a representative time snapshot, as shown in Fig. 17. The residuals converge to near-zero values upon training completion, demonstrating a reduction of over two orders of magnitude compared to its pre-training level.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

[1] R.L. Joesph, Ultrasonic Waves in Solid Media, Cambridge University Press, 2003.
[2] C.A. Chua, P. Cawley, P.B. Nagy, Scattering of the fundamental shear guided wave from a surface-breaking crack in plate-like structures, IEEE Trans. Ultrason. Ferroelectr. Freq. Control 66 (2019) 1887–1897, https://doi.org/10.1109/TUFFC.2019.2932227.
[3] P. Huthwaite, Guided wave tomography with an improved scattering model, Proc. Royal. Soc. A 472 (2016) 20160643, https://doi.org/10.1098/rspa.2016.0643.
[4] A. Spada, M. Capriotti, F. Lanza di Scalea, Global–local model for three-dimensional guided wave scattering with application to rail flaw detection, Struct. Health Monit. 21 (2) (2022) 370–386, https://doi.org/10.1177/14759217211000863.
[5] J.D. Achenbach, Wave Propagation in Elastic Solids, North-Holland, Amsterdam, 1973.
[6] B. Auld, Acoustic Fields and Waves in Solids, Krieger, Malabar, Florida, 1990.
[7] O. Diligent, T. Grahn, A. Boström, P. Cawley, M.J.S. Lowe, The low-frequency reflection and scattering of the S0 Lamb mode from a circular through-thickness hole in a plate: Finite element, analytical and experimental studies, J. Acoust. Soc. Am. 112 (6) (2002) 2589–2601, https://doi.org/10.1121/1.1512292.
[8] T. Grahn, Lamb wave scattering from a circular partly through-thickness hole in a plate, Wave Motion 37 (1) (2003) 63–80, https://doi.org/10.1016/S0165-2125(02)00051-3.
[9] L. Moreau, M. Caleap, A. Velichko, P.D. Wilcox, Scattering of guided waves by through-thickness cavities with irregular shapes, Wave Motion 48 (7) (2011) 586–602, https://doi.org/10.1016/j.wavemoti.2011.04.010.
[10] F.B. Cegla, A. Rohde, M. Veidt, Analytical prediction and experimental measurement for mode conversion and scattering of plate waves at non-symmetric circular blind holes in isotropic plates, Wave Motion 45 (3) (2008) 162–177, https://doi.org/10.1016/j.wavemoti.2007.05.005.
[11] A. Sedaghati, F. Honarvar, M. Tabatabaeipour, A.N. Sinclair, Investigation of the scattering of Lamb waves from a generalized circular cavity by using Poisson/Mindlin plate theories and numerical simulation, Proc. Inst. Mech. Eng. Part C J. Mech. Eng. Sci. 234 (2020) 152–170, https://doi.org/10.1177/0954406219871073.
[12] Z. Wang, Q. Leng, T. Li, Elastic wave scattering by flat-bottomed indentations on a plate, Int. J. Mech. Sci. 231 (2022) 107583, https://doi.org/10.1016/j.ijmecsci.2022.107583.
[13] L.R. Rose, C.H. Wang, Mindlin plate theory for damage detection: Imaging of flexural inhomogeneities, J. Acoust. Soc. Am. 127 (2) (2010) 754–763, https://doi.org/10.1121/1.3277217.
[14] B. Poddar, V. Giurgiutiu, Complex modes expansion with vector projection using power flow to simulate Lamb waves scattering from horizontal cracks and disbands, J. Acoust. Soc. Am. 140 (3) (2016) 2123–2133, https://doi.org/10.1121/1.4963087.
[15] B. Poddar, V. Giurgiutiu, Scattering of Lamb waves from a discontinuity: an improved: analytical approach, Wave Motion 65 (2016) 79–91, https://doi.org/10.1016/j.wavemoti.2016.03.009.
[16] M.F. Haider, B. Poddar, V. Giurgiutiu, Experimental validation of an analytical method to predict lamb wave scattering from a discontinuity, Smart Materi. Struct. 28 (1) (2018) 015012, https://doi.org/10.1088/1361-665X/aae910.
[17] M.F. Haider, R. Joseph, V. Giurgiutiu, B. Poddar, An efficient analytical global–local (AGL) analysis of the Lamb wave scattering problem for detecting a horizontal crack in a stiffened plate, Acta Mech. 231 (2020) 577–596, https://doi.org/10.1007/s00707-019-02555-z.
[18] M.F. Haider, M.Y. Bhuiyan, B. Poddar, B. Lin, V. Giurgiutiu, Analytical and experimental investigation of the interaction of Lamb waves in a stiffened aluminum plate with a horizontal crack at the root of the stiffener, J. Sound Vib. 431 (2018) 212–225, https://doi.org/10.1016/j.jsv.2018.06.018.
[19] Y. Shen, V. Giurgiutiu, WFR-2D: an analytical model for PWAS-generated 2D ultrasonic guided wave propagation, in: SPIE, 2014, https://doi.org/10.1117/12.2044798.
[20] Q. Deng, Z. Yang, Scattering of S0 Lamb mode in plate with multiple damage, Appl. Math. Model. 35 (1) (2011) 550–562, https://doi.org/10.1016/j.apm.2010.07.022.
[21] L. Sun, J. Wei, C. Peng, W. Hao, F. Teng, L. Fan, L. Zhang, Q. Sui, M. Jiang, Ultrasonic guided wave-based probabilistic diagnostic imaging method with Single-Path-Scattering sparse reconstruction for Multi-damage detection in composite structures, Mech. Syst. Sig. Process. 223 (2025) 111858, https://doi.org/10.1016/j.ymssp.2024.111858.
[22] C. Yeung, C.T. Ng, Time-domain spectral finite element method for analysis of torsional guided waves scattering and mode conversion by cracks in pipes, Mech. Syst. Sig. Process. 128 (2019) 305–317, https://doi.org/10.1016/j.ymssp.2019.04.013.
[23] H. Gravenkamp, J. Prager, A.A. Saputra, C. Song, The simulation of Lamb waves in a cracked plate using the scaled boundary finite element method, J. Acoust. Soc. Am. 132 (3) (2012) 1358–1367, https://doi.org/10.1121/1.4740478.
[24] J. Yue, P. Li, Numerical solution of the cavity scattering problem for flexural waves on thin plates: linear finite element methods, J. Comput. Phys. 497 (2024) 112606, https://doi.org/10.1016/j.jcp.2023.112606.
[25] Y. Cho, J.L. Rose, An elastodynamic hybrid boundary element study for elastic guided wave interactions with a surface breaking defect, Int. J. Solids Struct. 37 (30) (2000) 4103–4124, https://doi.org/10.1016/S0020-7683(99)00142-0.

[26] T. Maruyama, K. Kanda, S. Yamada, Three-dimensional hybrid SAFE-BEM for elastic guided-wave scattering in a plate with finite width, Eng. Anal. Bound. Elem. 163 (2024) 426–435, https://doi.org/10.1016/j.enganabound.2024.03.027.
[27] W. Shao, H. Sun, Q. Zhou, Y. Wang, X. Qing, A novel Lamb wave-based multi-damage dataset construction and quantification algorithm under the framework of multi-task deep learning, Struct. Health Monit. 23 (2) (2024) 1148–1169, https://doi.org/10.1177/14759217231185051.
[28] D. Ji, J. Lin, F. Gao, The high-sparsity Lamb wavefield reconstruction framework based on masked autoencoder, IEEE Trans. Instrum. Meas. 72 (2023) 1–10, https://doi.org/10.1109/TIM.2023.3316259.
[29] H. Zhang, J. Lin, J. Hua, T. Tong, Interpretable convolutional sparse coding method of Lamb waves for damage identification and localization, Struct. Health Monit. 21 (4) (2022) 1790–1804, https://doi.org/10.1177/14759217211044806.
[30] C. Humer, S. Höll, C. Kralovec, M. Schagerl, Damage identification using wave damage interaction coefficients predicted by deep neural networks, Ultrasonics 124 (2022) 106743, https://doi.org/10.1016/j.ultras.2022.106743.
[31] S. Zhang, C.M. Li, W. Ye, Damage localization in plate-like structures using time-varying feature and one-dimensional convolutional neural network, Mech. Syst. Sig. Process. 147 (2021) 107107, https://doi.org/10.1016/j.ymssp.2020.107107.
[32] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, J. Comput. Phys. 378 (2019) 686–707, https://doi.org/10.1016/j.jcp.2018.10.045.
[33] L. Yang, X. Meng, G.E. Karniadakis, B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data, J. Comput. Phys. 425 (2021) 109913, https://doi.org/10.1016/j.jcp.2020.109913.
[34] X. Meng, Z. Li, D. Zhang, G.E. Karniadakis, PPINN: Parareal physics-informed neural network for time-dependent PDEs, Comput. Methods Appl. Mech. Eng. 370 (2020) 113250, https://doi.org/10.1016/j.cma.2020.113250.
[35] L. Wang, H. Wang, L. Liang, J. Li, Z. Zeng, Y. Liu, Physics-informed neural networks for transcranial ultrasound wave propagation, Ultrasonics 132 (2023) 107026.
[36] H. Wang, J. Li, L. Wang, L. Liang, Z. Zeng, Y. Liu, On acoustic fields of complex scatters based on physics-informed neural networks, Ultrasonics 128 (2023) 106872, https://doi.org/10.1016/j.ultras.2022.106872.
[37] H. Sun, L. Peng, J. Lin, S. Wang, W. Zhao, S. Huang, Microcrack defect quantification using a focusing high-order SH guided wave EMAT: the physics-informed deep neural network GuwNet, IEEE Trans. Ind. Inf. 18 (5) (2021) 3235–3247, https://doi.org/10.1109/TII.2021.3105537.
[38] W. Zhou, Y.F. Xu, Damage identification for plate structures using physics-informed neural networks, Mech. Syst. Sig. Process. 209 (2024) 111111, https://doi.org/10.1016/j.ymssp.2024.111111.
[39] M.S. Go, H.K. Noh, J.H. Lim, Real-time full-field inference of displacement and stress from sparse local measurements using physics-informed neural networks, Mech. Syst. Sig. Process. 224 (2025) 112009, https://doi.org/10.1016/j.ymssp.2024.112009.
[40] L. Lu, P. Jin, G. Pang, Z. Zhang, G.E. Karniadakis, Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators, Nat. Mach. Intell. 3 (3) (2021) 218–229, https://doi.org/10.1038/s42256-021-00302-5.
[41] J. He, S. Koric, D. Abueidda, A. Najafi, I. Jasiuk, Geom-deeponet: a point-cloud-based deep operator network for field predictions on 3d parameterized geometries, Comput. Methods Appl. Mech. Eng. 429 (2024) 117130, https://doi.org/10.1016/j.cma.2024.117130.
[42] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, A. Anandkumar, Fourier neural operator for parametric partial differential equations, arXiv preprint arXiv:2010.08895 (2020). https://doi.org/10.48550/arXiv.2010.08895.
[43] C. Song, Y. Wang, High-frequency wavefield extrapolation using the Fourier neural operator, J. Geophys. Eng. 19 (2) (2022) 269–282, https://doi.org/10.1093/jge/gxac016.
[44] Z. Li, H. Zheng, N. Kovachki, D. Jin, H. Chen, B. Liu, K. Azizzadenesheli, A. Anandkumar, Physics-informed neural operator for learning partial differential equations, ACM/IMS J. Data Sci. 1 (3) (2024) 1–27, https://doi.org/10.1145/3648506.
[45] L. Wang, J. Li, S. Chen, Z. Fan, Z. Zeng, Y. Liu, Finite difference-embedded UNet for solving transcranial ultrasound frequency-domain wavefield, J. Acoust. Soc. Am. 155 (3) (2024) 2257–2269, https://doi.org/10.1121/10.0025391.
[46] J.D. Achenbach, Y. Xu, Wave motion in an isotropic elastic layer generated by a time-harmonic point load of arbitrary direction, J. Acoust. Soc. Am. 106 (1999) 83–90, https://doi.org/10.1121/1.427037.
[47] O. Ronneberger, P. Fischer, T. Brox, U-Net: convolutional networks for biomedical image segmentation, Med. Image Comput. Comput. Assist. Interv. (MICCAI) (2015), https://doi.org/10.1007/978-3-319-24574-4_28.
[48] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems, 2017, pp. 5998–6008.